Plain-language explanation: churn100-s4-c2

AI-assisted drafts for review. Nothing here has been verified by a person.

How to read this. Explanations are built up but read down: the summary first, then each part from the whole model down, in plain words, each sentence followed by the claims it rests on (C01, C02, ...; every cited claim was computed on the model and checked). Numbers are changes in predicted risk for a typical patient. The technical record (original hypotheses, formulas, evidence tables with the test check, ranges and n) is at the end. The text describes what the model does, not the world; whether it is plausible is for the reviewer to judge.

What changed in v5e (figures). v5e: every diagram is a numbered figure with a caption (list of figures after the contents) and the text refers to it; the whole-network figure has its own landscape page, scaled to the page, with a short caption and its description in the text; an input used by several parts is drawn under each of them (same input, repeated for readability, dotted outline); and an appendix figure shows the trained network before any explanation (no parts, no splits).

What changed in v5c (tables and diagrams). v5c: the evidence tables are landscape, compact and shaded by row, with the header repeated on every page (longtable and pdflscape, used in this review PDF only; the thesis preamble has neither). In the diagrams each input is drawn once, in a shared row along the bottom (the explanation's split copies of an input are listed in the technical detail instead), every cluster has one output rectangle with its plain name, the inputs that feed the output directly meet in a drawn 'sum', and a one-line legend explains the edges.

What changed in v5b (diagrams). v5b uses one grouping everywhere, the annotation hierarchy: the summary's diagram is the same picture as the whole model's section, and a part that is itself a weighted sum (blood pressure with cholesterol) is a part of its own, described as one that adds up, so the text counts three parts throughout; the lint checks every 'N parts' against the hierarchy. A full-structure diagram follows the summary: every node and connection, nested by part, with the inputs that feed the output directly in their own cluster and the unused input columns listed. Each section from the top down adds its own region of that diagram, with stubs for connections from or to other parts; each connection is drawn in exactly one section. The gate's text now says that it lowers the risk for the patients it acts on, and that more vessels and higher cholesterol take that lowering away; its vessel statements are one per group.

What changed in v5 (read down). v5 reads the explanation down. The summary (one line, a diagram, one paragraph per block) is as in v4, except that the paragraph on what adds up is split into protective items, risk-raising items and rare results, and an input that also feeds a part is quoted with its direct contribution only (the rest is in that part's paragraph). Then the document walks the annotation hierarchy from the whole model down: for each part a plain hypothesis (what it does for the prediction), plain evidence (what was checked and what it showed; for a part made of others, what they are and how they combine), a diagram one level down and where to read on. The original hypotheses, closed forms and evidence tables are in 'Technical detail' at the end. Statements about structure (what a part is made of, a copy of another part, a formula that reproduces it) are now claims of their own, checked on the graph and the data.

v4. v4 follows the structure of the explanation. The summary opens with one line and a diagram at the same grouping, then gives one paragraph per top-level block of the annotation hierarchy: one for everything that adds up independently (the inputs that feed the model score directly, and parts that are a weighted sum), and one for each part that combines its inputs in another way: what it combines, how, and for whom. 'Parts in detail' follows the hierarchy below that, with a diagram one level down for each part that contains others. The prose no longer says 'clearly', 'slightly', 'consistently' or 'on the training data only': the numbers carry it, and the test check, the ranges and n stay in the evidence tables. Only genuinely small groups are marked, with their n. The lint rejects the confidence words and checks the block structure.

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53 The network as trained, before any explanation, in full. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. Not used (no connection): Alaska, Alabama, Iowa, Idaho, Illinois, Kansas, Maryland, Maine, Michigan, Minnesota, Missouri, Mississippi, Montana, North Carolina, Nebraska, New Hampshire, New Mexico, Nevada, New York, Ohio, California, South Carolina, Texas, Utah, Wisconsin, Colorado, Wyoming, Washington DC, Delaware, area code 408, area code 510, total evening calls, total night-time minutes, total night-time calls, total night-time charge, total international calls, total international charge. . 55

1 churn100-s4-c2

Genome `4206744f-b5ce-4488-bc53-a39aa08fe972`, explanation `33d6db5d-fbdf-41c8-a597-fc997715b92f`. Split: train primary (n=4000), test as a check (n=1000). Claims: 113, verified 110, not supported 3.
AI-assisted draft for review (unverified)

Reference (typical) customer: the typical customer: the median training customer for every number and the most common training category for every category (not a real customer). account length: 100; area code: area code 415; international plan: no international plan; number customer service calls: 1 calls; number vmail messages: 0 messages; state: West Virginia; total day calls: 100 calls; total day charge: 30.69 US dollars; total day minutes: 180 minutes; total eve calls: 100 calls; total eve charge: 17.12 US dollars; total eve minutes: 201 minutes; total intl calls: 4 calls; total intl charge: 2.78 US dollars; total intl minutes: 10 minutes; total night calls: 100 calls; total night charge: 9.02 US dollars; total night minutes: 200 minutes; voice mail plan: no voice-mail plan. Predicted chance that a customer leaves: 3%.

1.1 Summary

For a typical customer (in West Virginia, area code 415, no international or voice-mail plan, no voice-mail messages, 1 call to customer service and median usage) the model predicts a chance of leaving of about 3%; it adds up six parts: fixed-weight inputs, service calls, heavy daytime use, state units, an evening unit and a small voice-mail unit. [C046] Figure 1 shows the model at this level: one box per paragraph below.

How to read the diagrams. Edge colour is the sign of that one connection: blue positive, red negative; thicker is a larger weight. The outline and arrow on an input (and on a part's output box) is its overall effect on the final prediction, read from the verified claims: orange and up arrow raises it, teal and down arrow lowers it, grey and double arrow depends (it goes both ways, or differs between levels). A negative connection into a part that itself raises the prediction means that input lowers the prediction; through a hinge the overall effect can only be read from the claims, not from the edges. Up arrow means towards churn; down arrow, away from it. A double outline marks an input (or a part's output) that acts at a fixed rate everywhere: it adds the same amount to the score per unit, so it is explained by its rate alone (see 'Linear factors'); a dashed double outline, at a fixed rate on the observed data. The normal outline is an input whose effect switches on and off somewhere. For example, a voice-mail plan has a negative connection into 'Fixed-weight inputs', a part that raises the predicted chance that a customer leaves, so a voice-mail plan lowers it.

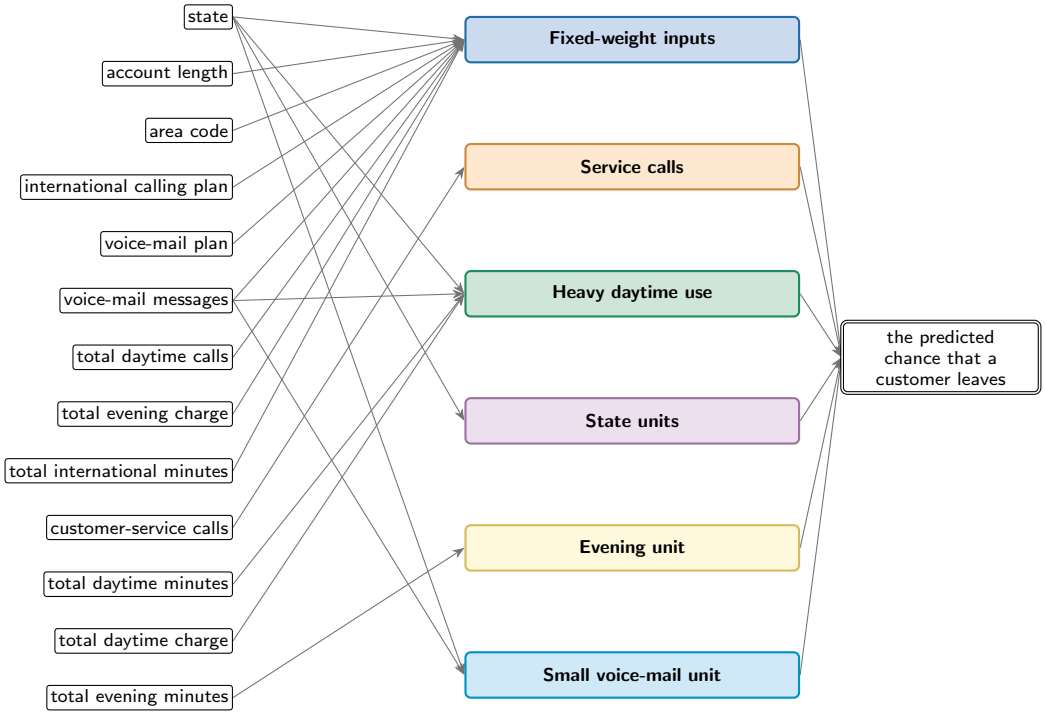


Figure 1: The model at the level of the summary: one box per part.

Fixed-weight inputs. This part adds up most inputs with fixed weights: an international plan raises a typical customer's chance of leaving from about 3% to about 31%, and a voice-mail plan lowers it to under 1%. [C047, C051] More voice-mail messages raise it steadily, from about 3% at 0 to about 10% at 32 messages, flattening out near the bottom because the predicted chance cannot go much lower. [C049] A higher evening charge and more

international minutes also raise it steadily, each from about 2% to about 4% across their usual range. [C055, C056] Through it, the state moves the chance from about 3% in West Virginia to about 11% in Washington and about 7% in New Jersey, and to under 1% in Hawaii and in Rhode Island (72 to 122 training customers per state). [C048, C050, C052, C053]

Service calls. This part adds up two parts: a rise from 3 customer-service calls upwards and a second rise from 4 calls upwards. [C079] From 3 calls upwards it adds to the chance slowly at first, then faster: for a typical such customer, 4 calls take it from about 3% to about 18%. [C078] Up to 3 calls it barely moves a typical customer's chance, which stays at about 3%, rising slowly at first, then faster. [C077]

Heavy daytime use. Daytime minutes add to the chance only once they rise above about 221 minutes (between 186 and 221 minutes for different customers), in an uneven way: for a typical such customer, from about 3% at 0 minutes to about 19% at 271 minutes. [C088] More voice-mail messages lower it in an uneven way: for the typical customer it acts on, from about 12% at 0 to about 6% at 32 messages. [C080] Customers in Washington get less from it: about 7% instead of about 12% for the typical customer it acts on. [C082]

State units. This part reads only the state: it raises a typical customer's chance from about 3% to about 22% in Rhode Island, and by less than a point in Indiana. [C100, C101]

Evening unit. This part reads evening minutes: nothing until about 217 minutes, then steadily, but a typical customer's chance stays at about 3%. [C103, C104]

Small voice-mail unit. This small part reads voice-mail messages and being in Louisiana, and barely changes the predictions (by at most one point). [C111]

The whole network. Figure 2 (next page, landscape) draws every unit and connection of the explained model, nested by part: shaded clusters are the parts of the model and darker boxes the parts inside them; circles are hidden units (h1, h2, ...); each cluster's rectangle is its output, and a dashed 'sum' stands for an addition the next unit does (drawn, not a unit of the model); the hexagon is the model's output. An input used by several parts is drawn under each of them (same label, dotted outline, the first copy marked with the number of copies): it is the same input, repeated for readability. Not used by the model (input columns with no connection): state: 29 of 51 levels, area code 408, area code 510, total evening calls, total night-time minutes, total night-time calls, total night-time charge, total international calls, total international charge.

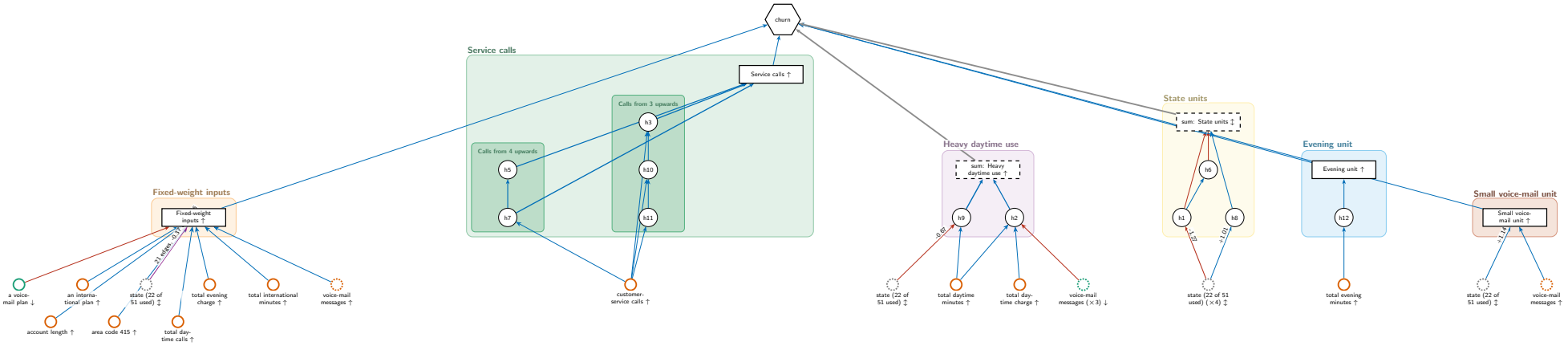


Figure 2: Every unit and connection of the explained model, nested by part. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Worth checking against what is known. Worth checking: for customers in Rhode Island the fixed-weight inputs lower the chance to under 1% while the state units raise it to about 22%, so overall it stays at about 3%. [C053, C100, C033] Worth checking: the state moves the chance by several points (Washington about 11%, New Jersey about 7%, Hawaii under 1%), based on 72 to 122 training customers per state. [C002, C005, C007] Worth checking: the model treats a voice-mail plan as lowering the chance of leaving but more voice-mail messages as raising it. [C006, C003] Worth checking: voice-mail messages raise the chance through the fixed-weight inputs but lower it through heavy daytime use. [C049, C080]

1.2 The model, read from the top down

Each part of the explanation in turn, from the whole model down to the smallest parts: what it does for the prediction (hypothesis), what was checked (evidence), a diagram one level down, and where to read on. The technical record of each part is in "Technical detail" at the end.

1.2.1 The whole model

Figure 3 shows what feeds this part; Figure 4 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

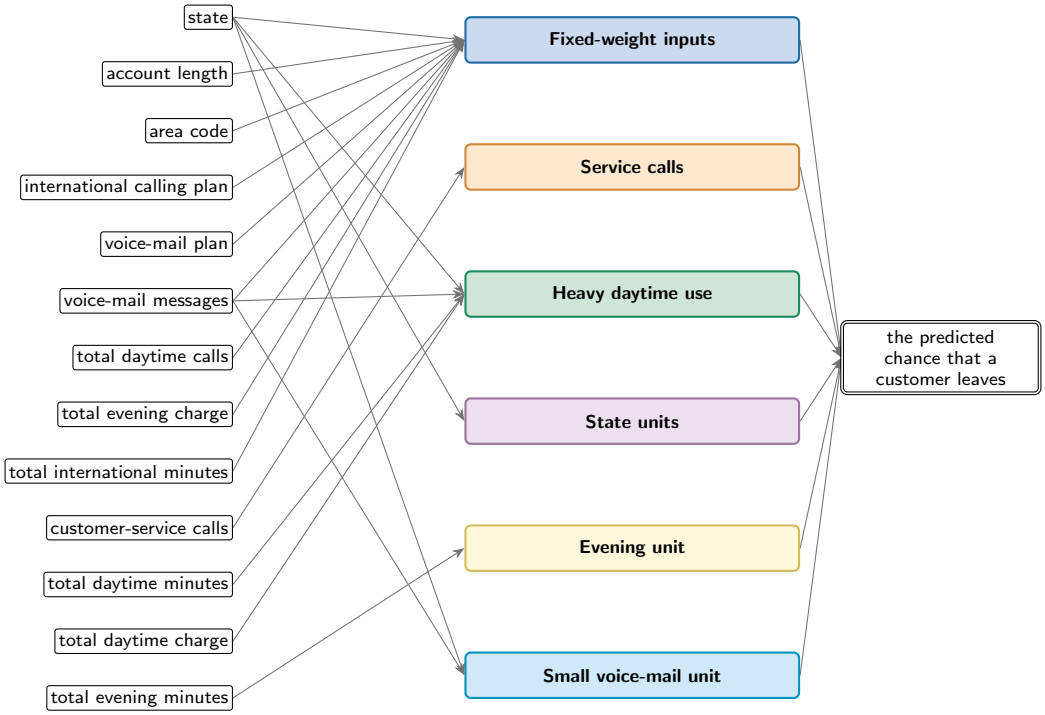


Figure 3: The whole model: what feeds it, one level down.

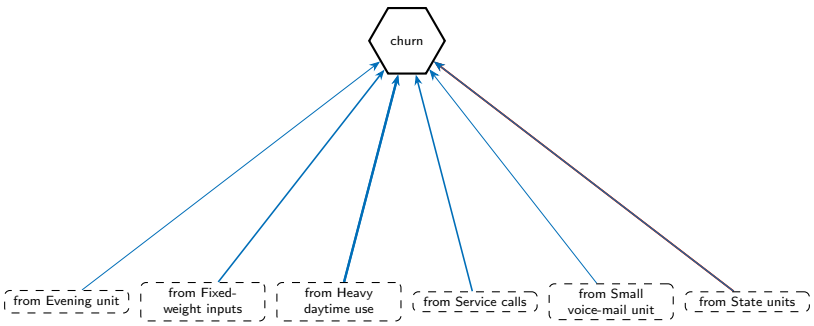


Figure 4: The whole model: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. The model adds up six parts into one score: fixed-weight inputs, service calls, heavy daytime use, state units, an evening unit and a small voice-mail unit. [C046]

Evidence. It is made up of these six parts (each explained below); the score is their sum, which was checked on all 4000 training customers. [C046] Changing one input at a time across the whole model, an international plan takes a typical customer's chance of leaving from about 3% to about 31%, and a voice-mail plan lowers it to under 1%. [C001, C006] More daytime minutes raise it in an uneven way, from about 3% at 112 minutes to about 8% at 249 minutes, flattening out near the bottom because the predicted chance cannot go much lower. [C004] In Washington the chance is about 11%, against about 3% in West Virginia. [C002] The model ignores evening calls, international calls, the international charge, night-time calls, the night-time charge and night-time minutes. [C040, C041, C042, C043, C044, C045]

Evidence plots: Figure 5, Figure 6.

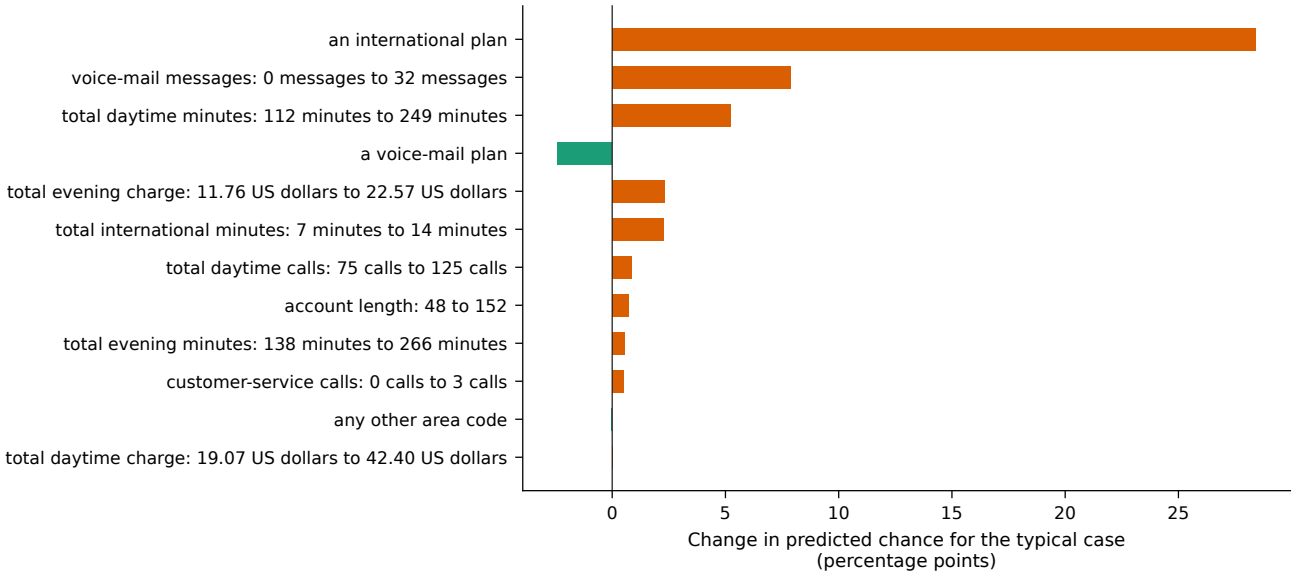


Figure 5: The whole model: Each input's change in the predicted value for the typical case across the whole model, largest first (orange: raises it, teal: lowers it).

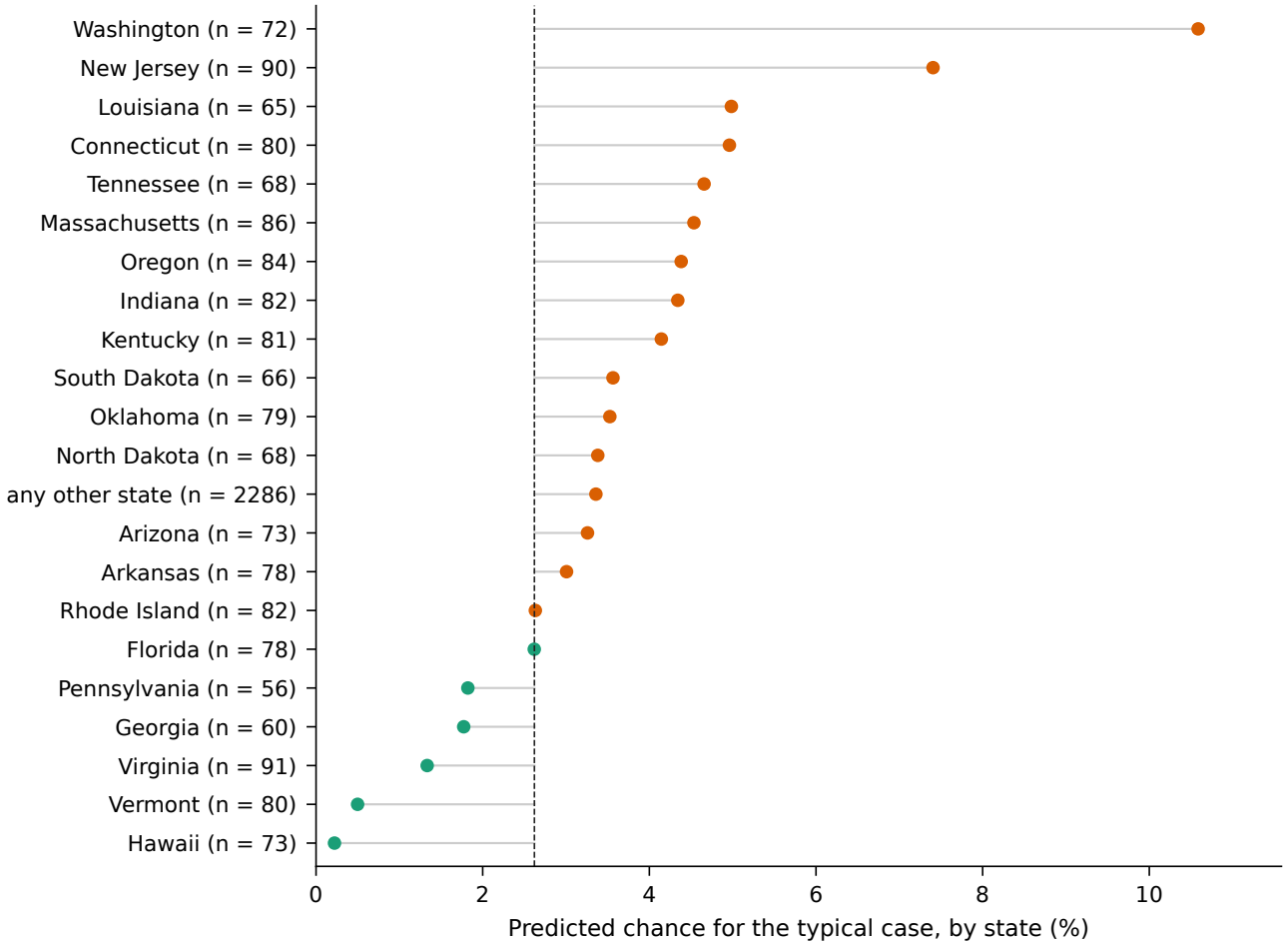


Figure 6: The whole model: Every level of state the model reads: the predicted value for the typical case at that level (dashed: the reference level); n = training customers; grey: few customers or not supported.

Dive deeper: Fixed-weight inputs (1.2.2); Service calls (1.2.3); Heavy daytime use (1.2.6); State units (1.2.7); Evening unit (1.2.8); Small voice-mail unit (1.2.9). Technical detail: T.1.

1.2.2 Fixed-weight inputs

Part of: The whole model (1.2.1).
Figure 7 shows what feeds this part; Figure 8 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

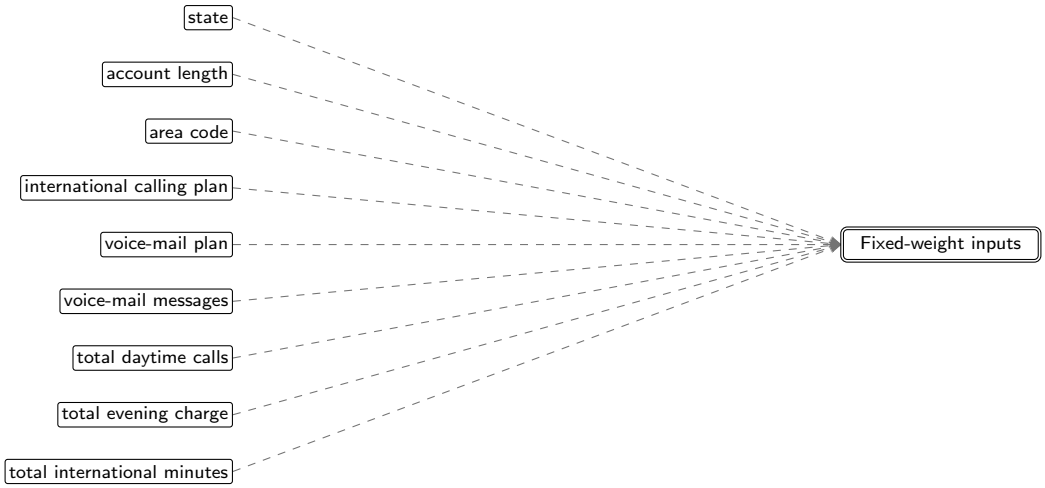


Figure 7: Fixed-weight inputs: what feeds it, one level down.

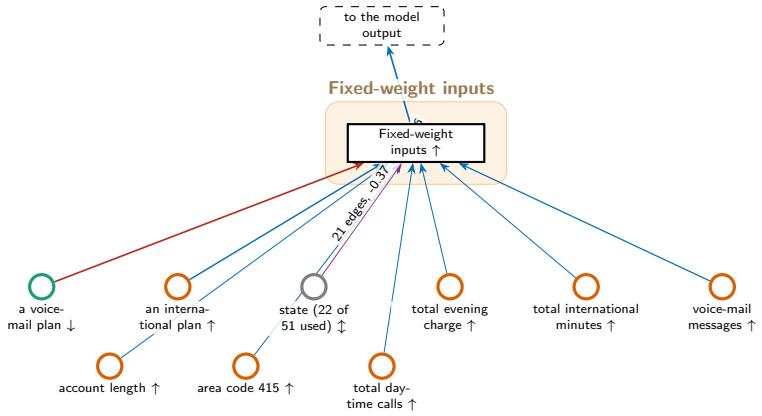


Figure 8: Fixed-weight inputs: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part adds up most inputs with fixed weights: an international plan, voice-mail messages, the evening charge and international minutes raise the chance of leaving, steadily for the numbers, and a voice-mail plan lowers it. [C047, C049, C055, C056, C051]

Evidence. Through it, an international plan takes a typical customer from about 3% to about 31%, and a voice-mail plan lowers the chance to under 1%. [C047, C051] More voice-mail messages raise it steadily, from about 3% at 0 to about 10% at 32 messages, flattening out near the bottom because the predicted chance cannot go much lower. [C049] Through it, the state moves the chance from about 3% in West Virginia to about 11% in Washington and about 7% in New Jersey, and to under 1% in Hawaii and in Rhode Island (72 to 122 training customers per state). [C048, C050, C052, C053] Its formula reproduces the part exactly on all 4000 training customers. [C076]
Evidence plots: Figure 9, Figure 10, Figure 11, Figure 12.

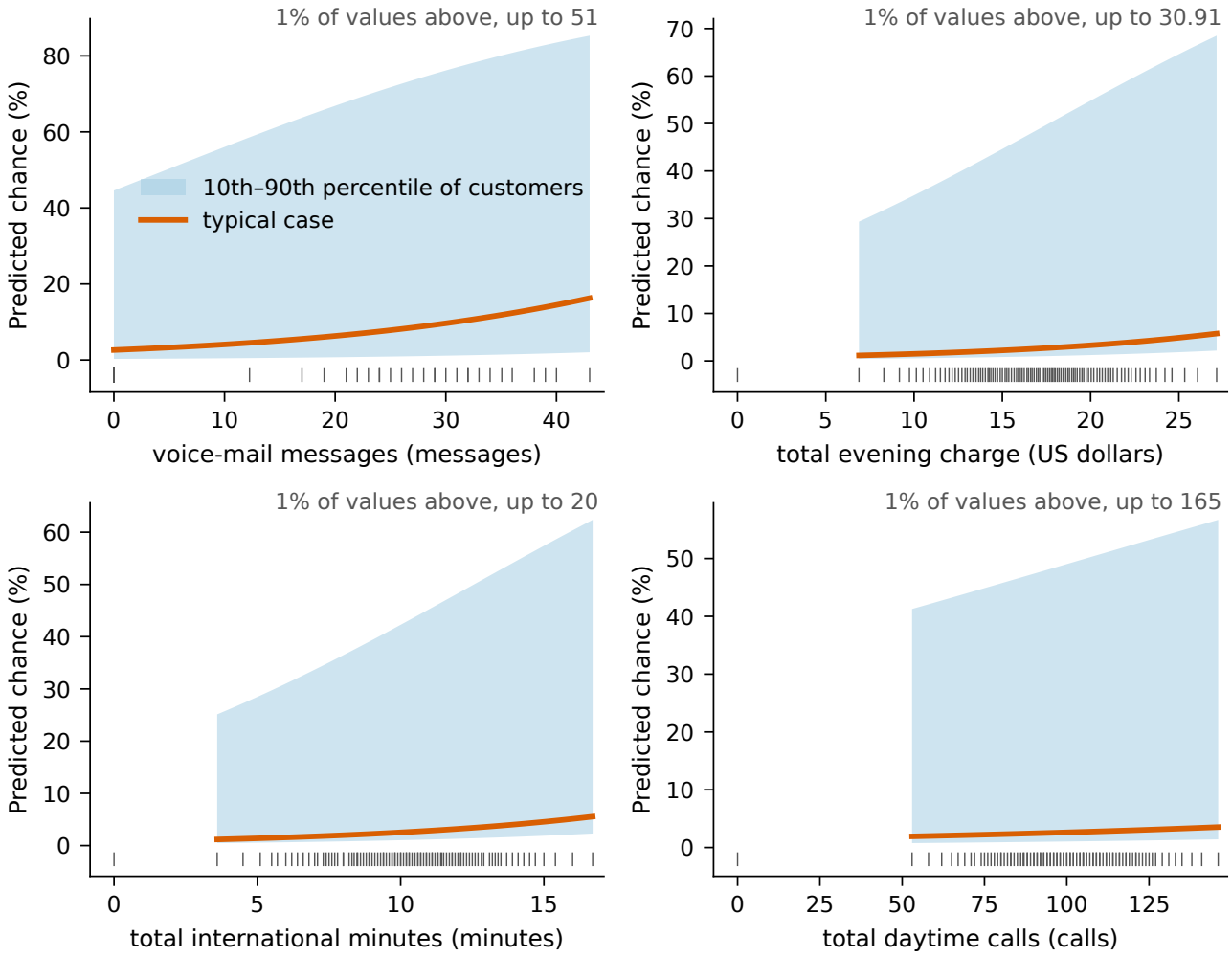


Figure 9: Fixed-weight inputs: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

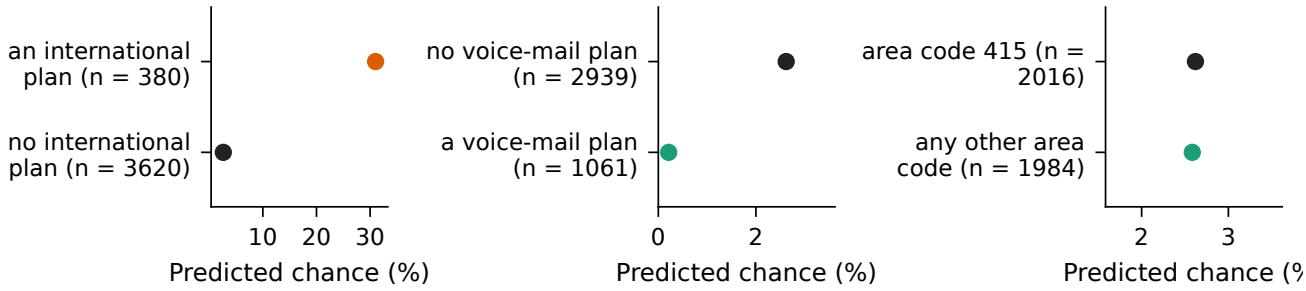


Figure 10: Fixed-weight inputs: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

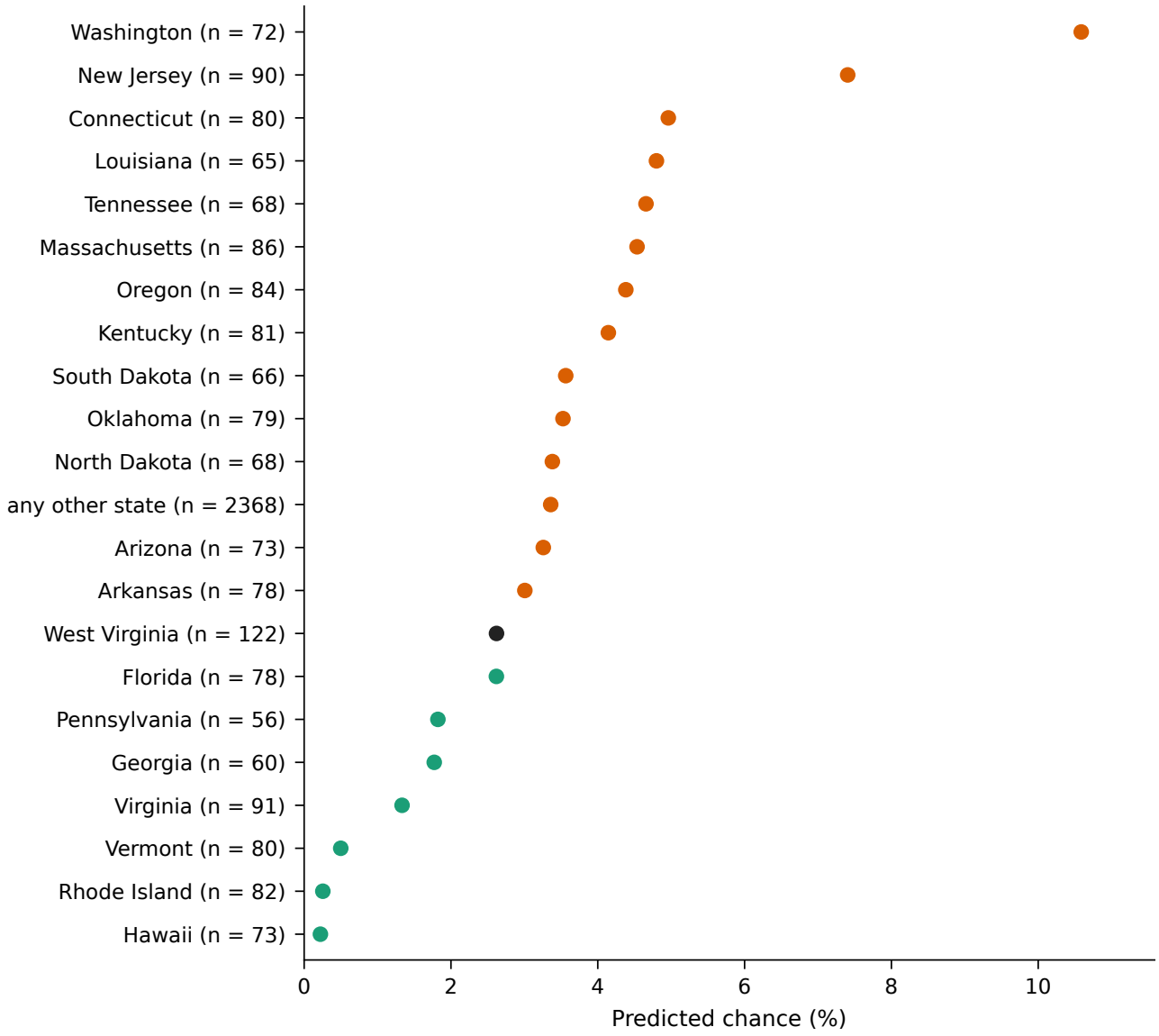


Figure 11: Fixed-weight inputs: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

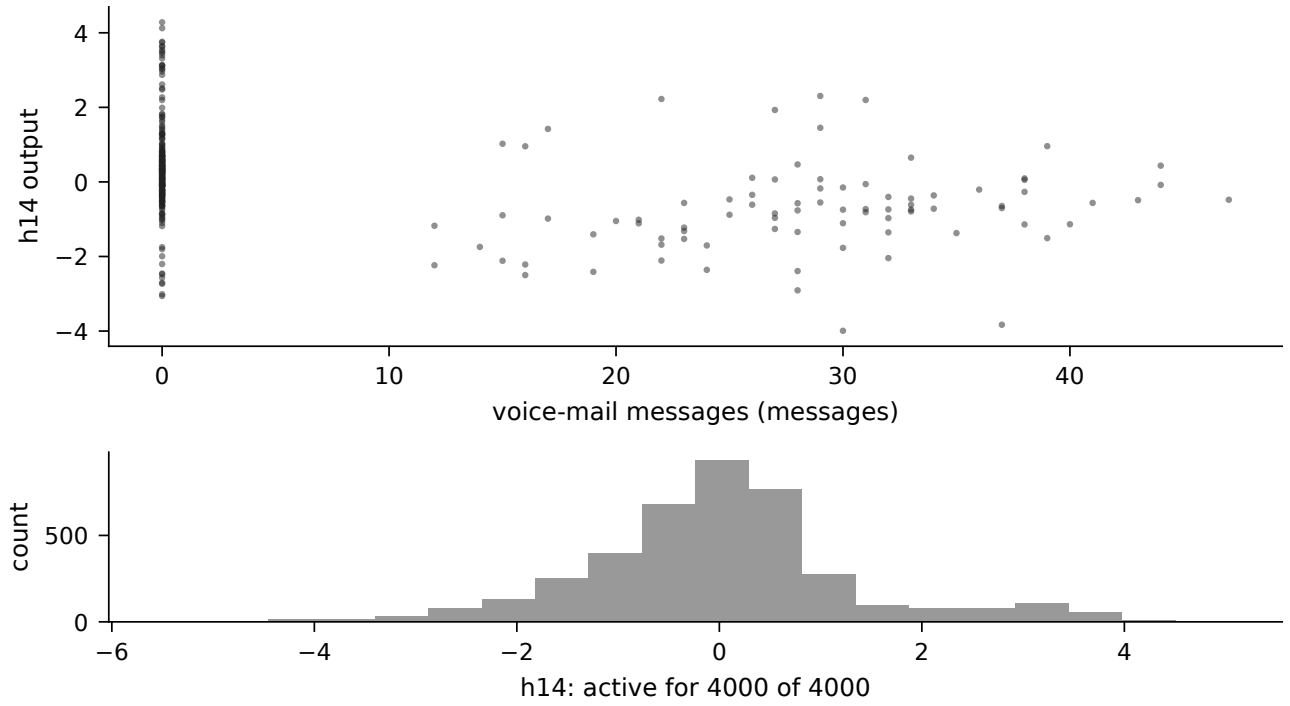


Figure 12: Fixed-weight inputs: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.2.

1.2.3 Service calls

Part of: The whole model (1.2.1).
Figure 13 shows what feeds this part; Figure 14 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 13: Service calls: what feeds it, one level down.

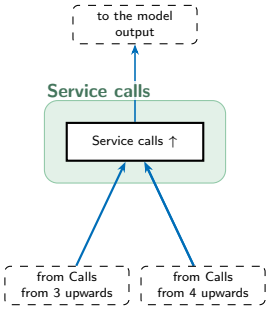


Figure 14: Service calls: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the chance of leaving with calls to customer service from 3 calls upwards, slowly at first, then faster; it adds up two parts. [C079, C078, C077]

Evidence. It is made of two parts that are added: calls from 3 upwards and calls from 4 upwards. [C079] From 3 calls upwards it adds to the chance slowly at first, then faster: for a typical such customer, 4 calls take it from

about 3% to about 18%. [C078] Up to 3 calls it barely moves a typical customer's chance, which stays at about 3%, rising slowly at first, then faster. [C077]
Evidence plots: Figure 15, Figure 16, Figure 17.
Figure 16 shows this part's own output, unit h15: it switches on at about 2.899 calls of customer-service calls, and is zero below.

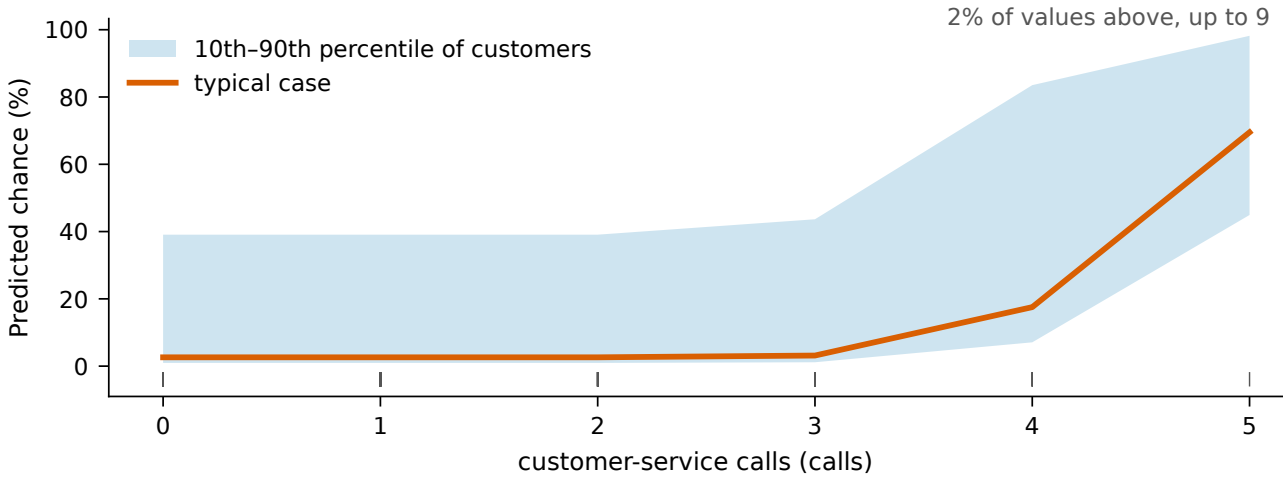


Figure 15: Service calls: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

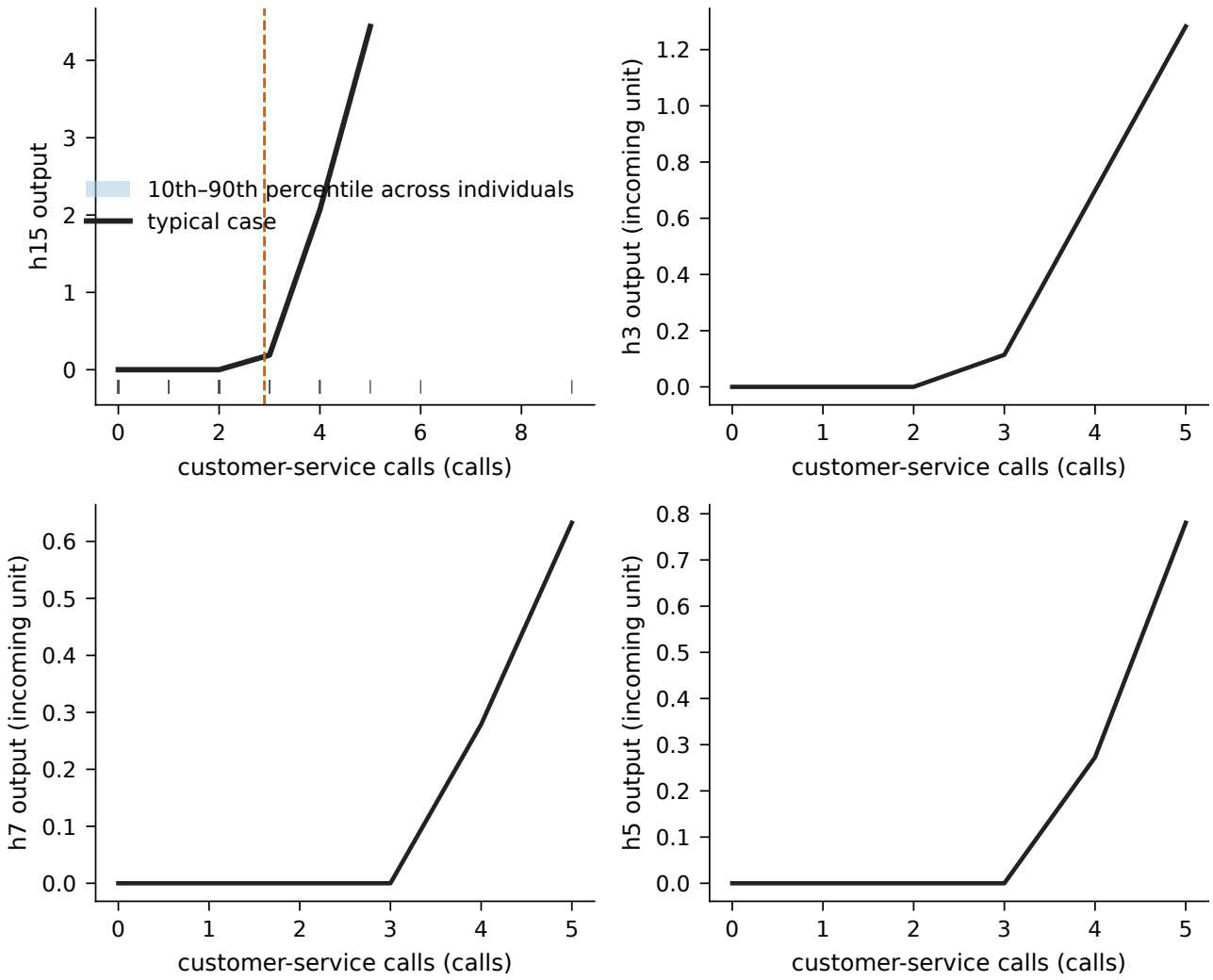


Figure 16: Service calls: This part's own output, unit h15 (combines 3 non-linear units), on its own scale (its activation), with every other input at the typical case; it is active for 848 of 4000. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

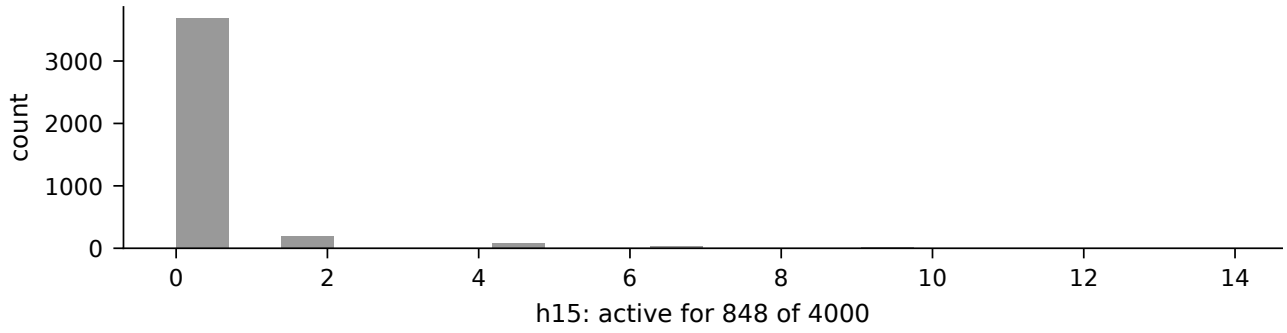


Figure 17: Service calls: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Dive deeper: Calls from 4 upwards (1.2.4); Calls from 3 upwards (1.2.5). Technical detail: T.3.

1.2.4 Calls from 4 upwards

Part of: Service calls (1.2.3).
Figure 18 shows what feeds this part; Figure 19 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 18: Calls from 4 upwards: what feeds it, one level down.

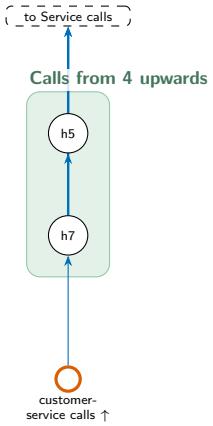


Figure 19: Calls from 4 upwards: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the chance of leaving from 4 customer-service calls upwards, slowly at first, then faster. [C093]

Evidence. It acts on 318 of 4000 training customers, 159 of whom left; for the typical customer it acts on, the chance is about 8% without it and about 18% with it. [C094] Through it, the chance rises slowly at first, then faster: 4 calls take a typical such customer from about 3% to about 6%. [C093] Its formula reproduces the part exactly on all 4000 training customers. [C095]
Evidence plots: Figure 20, Figure 21, Figure 22.
Figure 20 shows this part's own output, unit h7: it switches on at about 3.212 calls of customer-service calls, and is zero below.
Figure 21 shows this part's own output, unit h5: it switches on at about 3.463 calls of customer-service calls, and is zero below.

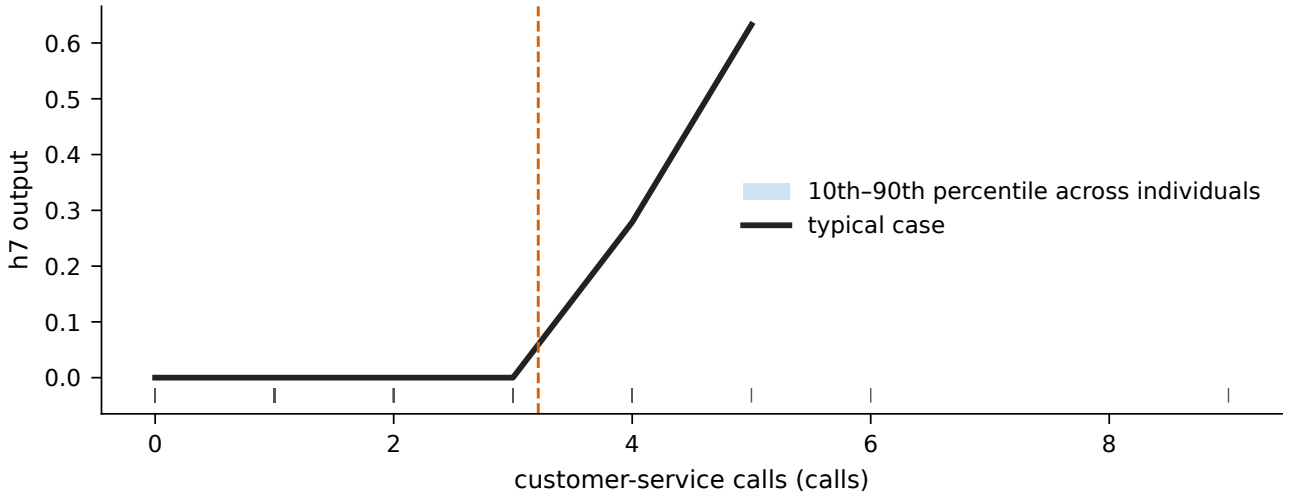


Figure 20: Calls from 4 upwards: This part's own output, unit h7 (the output of 'Calls from 4 upwards'), on its own scale (its activation), with every other input at the typical case; it is active for 318 of 4000. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

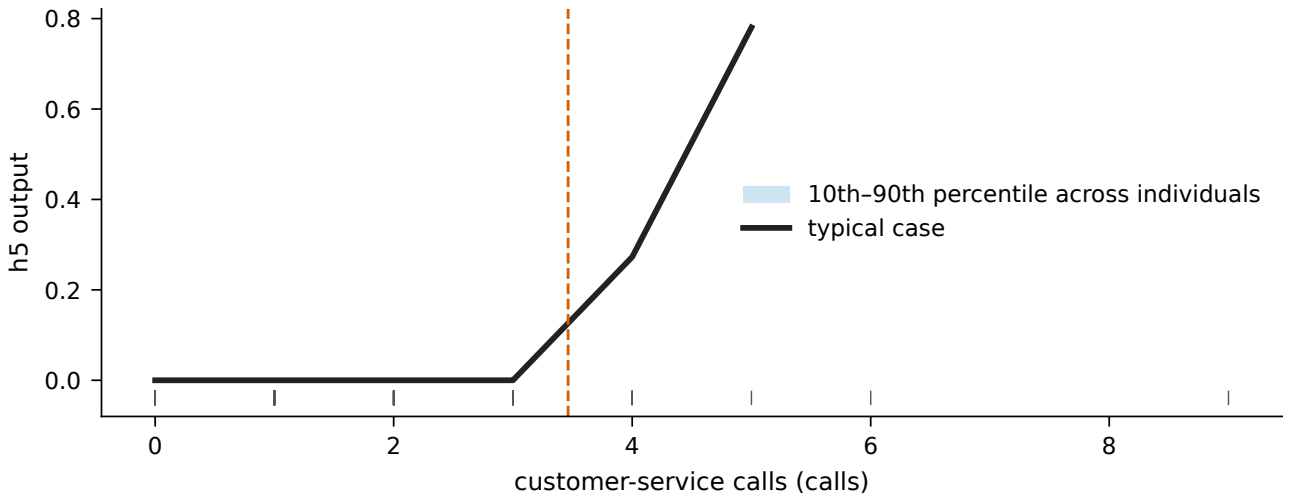


Figure 21: Calls from 4 upwards: This part's own output, unit h5 (the output of 'Calls from 4 upwards'), on its own scale (its activation), with every other input at the typical case; it is active for 318 of 4000. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

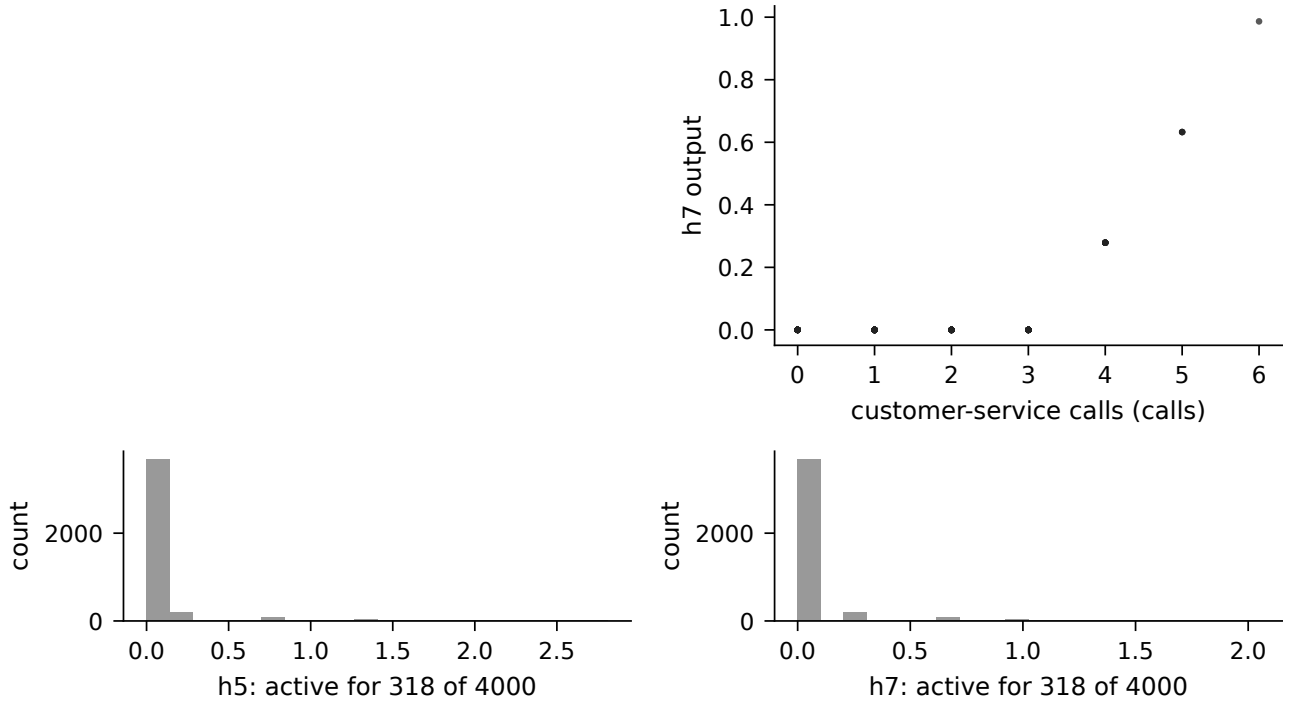


Figure 22: Calls from 4 upwards: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.4.

1.2.5 Calls from 3 upwards

Part of: Service calls (1.2.3).
Figure 23 shows what feeds this part; Figure 24 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

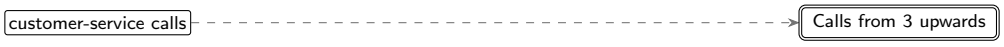


Figure 23: Calls from 3 upwards: what feeds it, one level down.

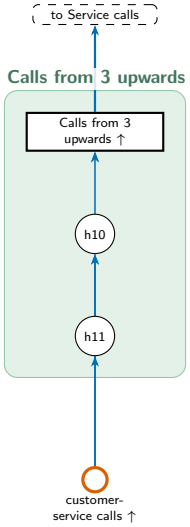


Figure 24: Calls from 3 upwards: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the chance of leaving from 3 customer-service calls upwards, slowly at first, then faster. [C097, C096]

Evidence. It acts on 848 of 4000 training customers, 215 of whom left; for the typical customer it acts on, the chance is about 3% without it and about 4% with it. [C098] Through it, the chance rises slowly at first, then faster: 4 calls take a typical such customer from about 3% to about 8%. [C097] Its formula reproduces the part exactly on all 4000 training customers. [C099]
Evidence plots: Figure 25, Figure 26, Figure 27.
Figure 26 shows this part's own output, unit h3: it switches on at about 2.805 calls of customer-service calls, and is zero below.

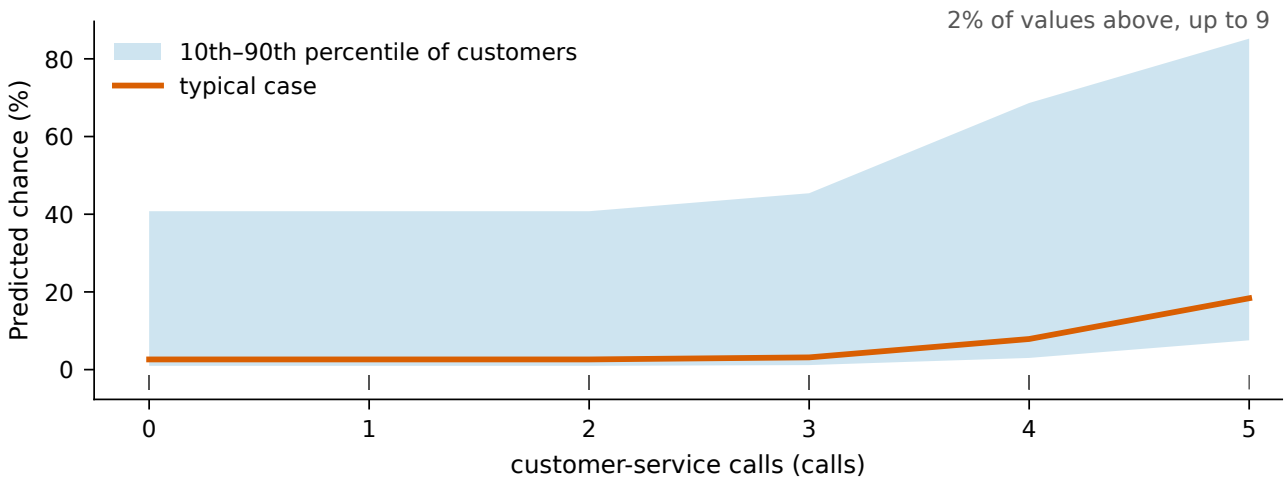


Figure 25: Calls from 3 upwards: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

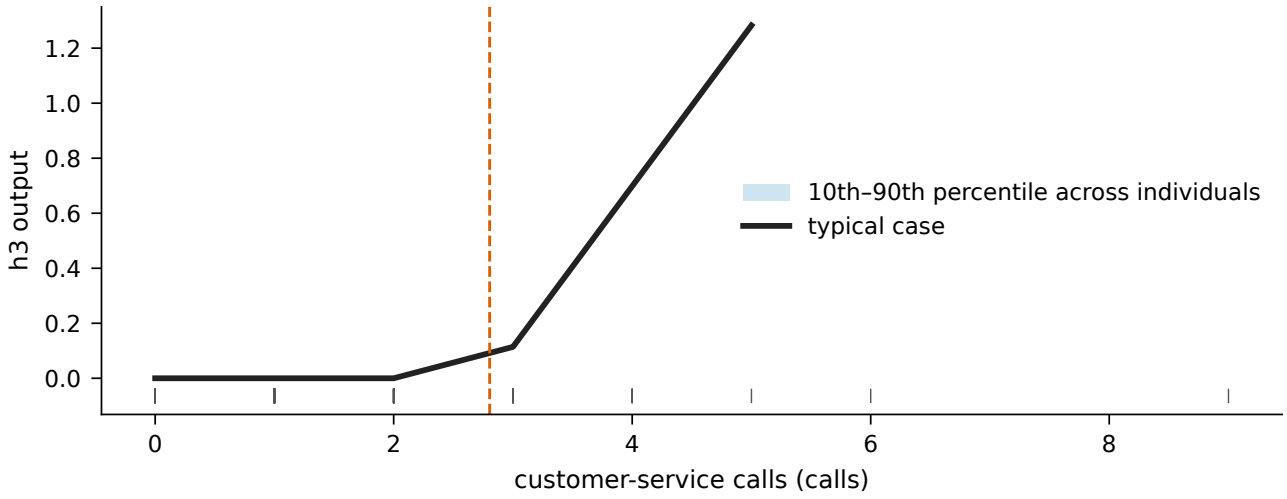


Figure 26: Calls from 3 upwards: This part's own output, unit h3 (the output of 'Calls from 3 upwards'), on its own scale (its activation), with every other input at the typical case; it is active for 848 of 4000. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

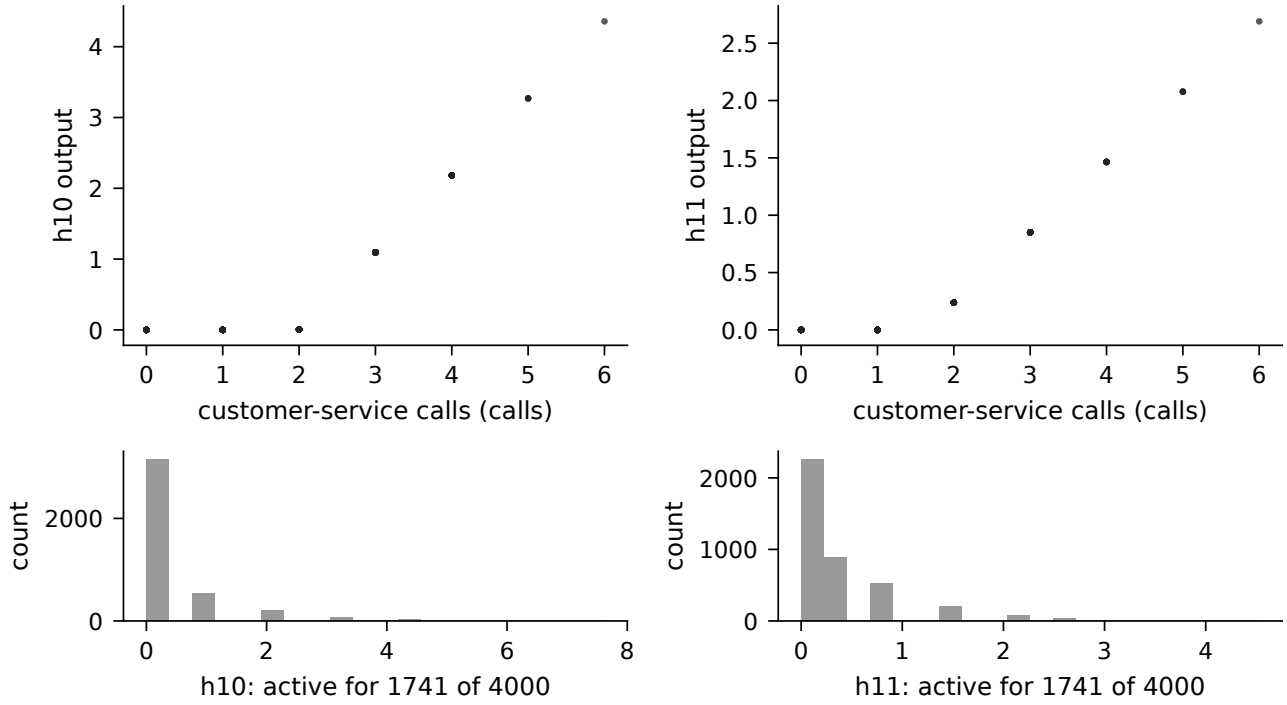


Figure 27: Calls from 3 upwards: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.5.

1.2.6 Heavy daytime use

Part of: The whole model (1.2.1).
Figure 28 shows what feeds this part; Figure 29 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

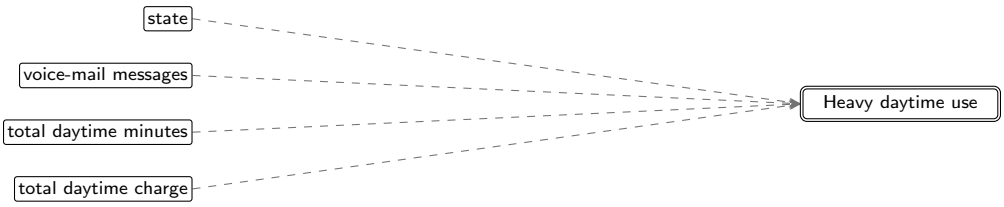


Figure 28: Heavy daytime use: what feeds it, one level down.

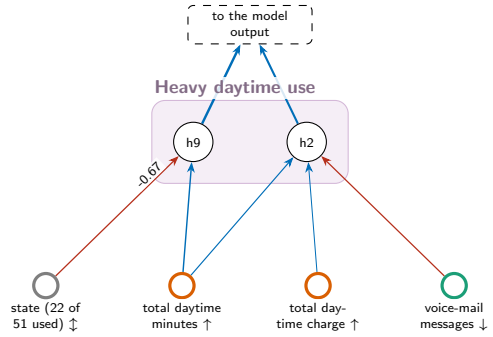


Figure 29: Heavy daytime use: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the chance of leaving for heavy daytime users: daytime minutes add to it only above about 221 minutes (between 186 and 221 minutes for different customers), in an uneven way, and voice-mail messages and being in Washington lower it. [C088, C080, C082]

Evidence. This part acts on 1099 of 4000 training customers, 291 of whom left; for the typical customer it acts on, the chance is about 3% without it and about 12% with it. [C091] Daytime minutes add to the chance only once they rise above about 221 minutes (between 186 and 221 minutes for different customers), in an uneven way: for a typical such customer, from about 3% at 0 minutes to about 19% at 271 minutes. [C088] More voice-mail messages lower it in an uneven way: for the typical customer it acts on, from about 12% at 0 to about 6% at 32 messages. [C080] Customers in Washington get less from it: about 7% instead of about 12% for the typical customer it acts on. [C082] Its formula reproduces the part exactly on all 4000 training customers. [C092]
Evidence plots: Figure 30, Figure 31, Figure 32, Figure 33, Figure 34, Figure 35, Figure 36, Figure 37, Figure 38.
Figure 36 shows this part's own output, unit h2: over its two main inputs, with the line where it switches on.
Figure 37 shows this part's own output, unit h9: it switches on at about 221.1 minutes of total daytime minutes, and is zero below.

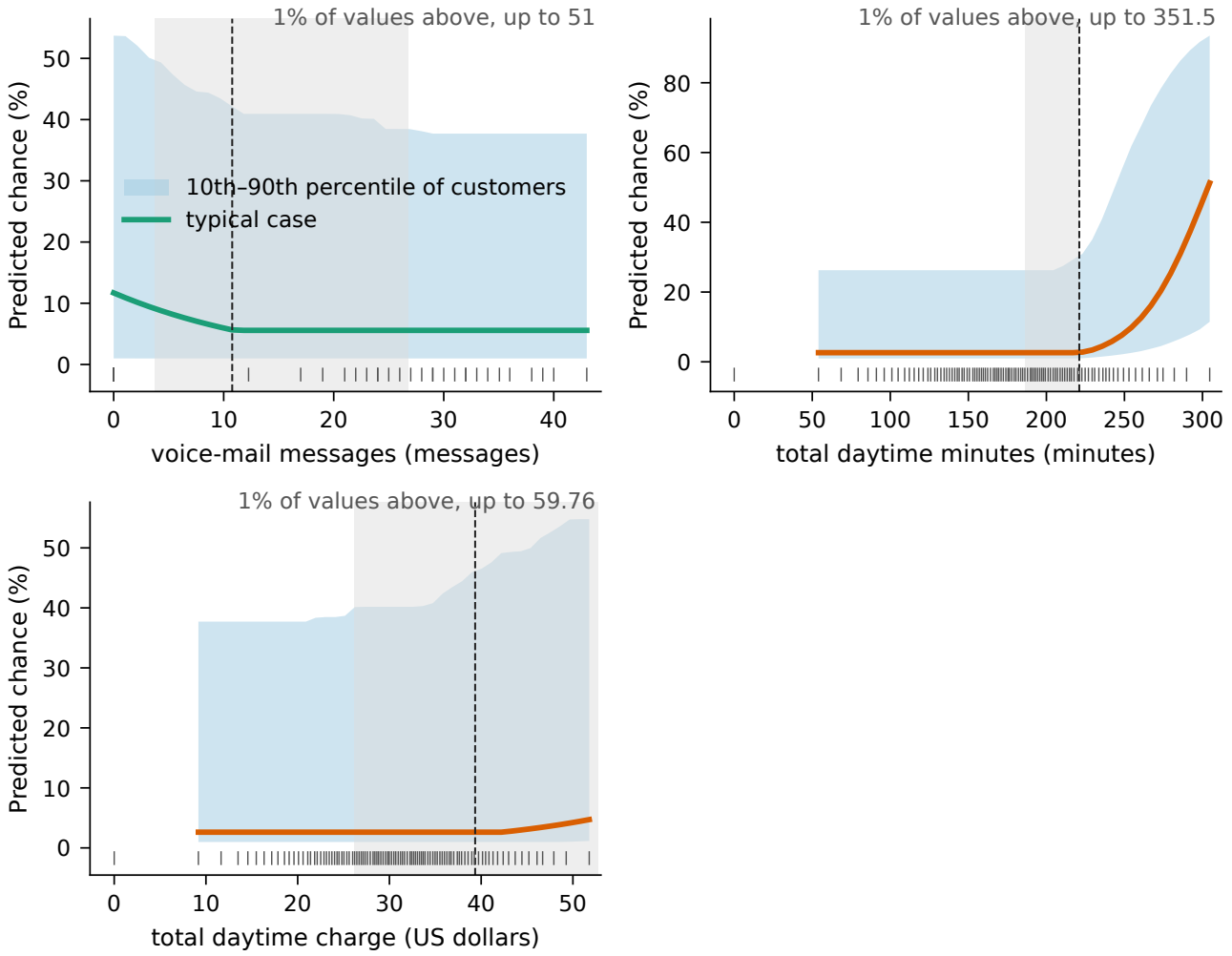


Figure 30: Heavy daytime use: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

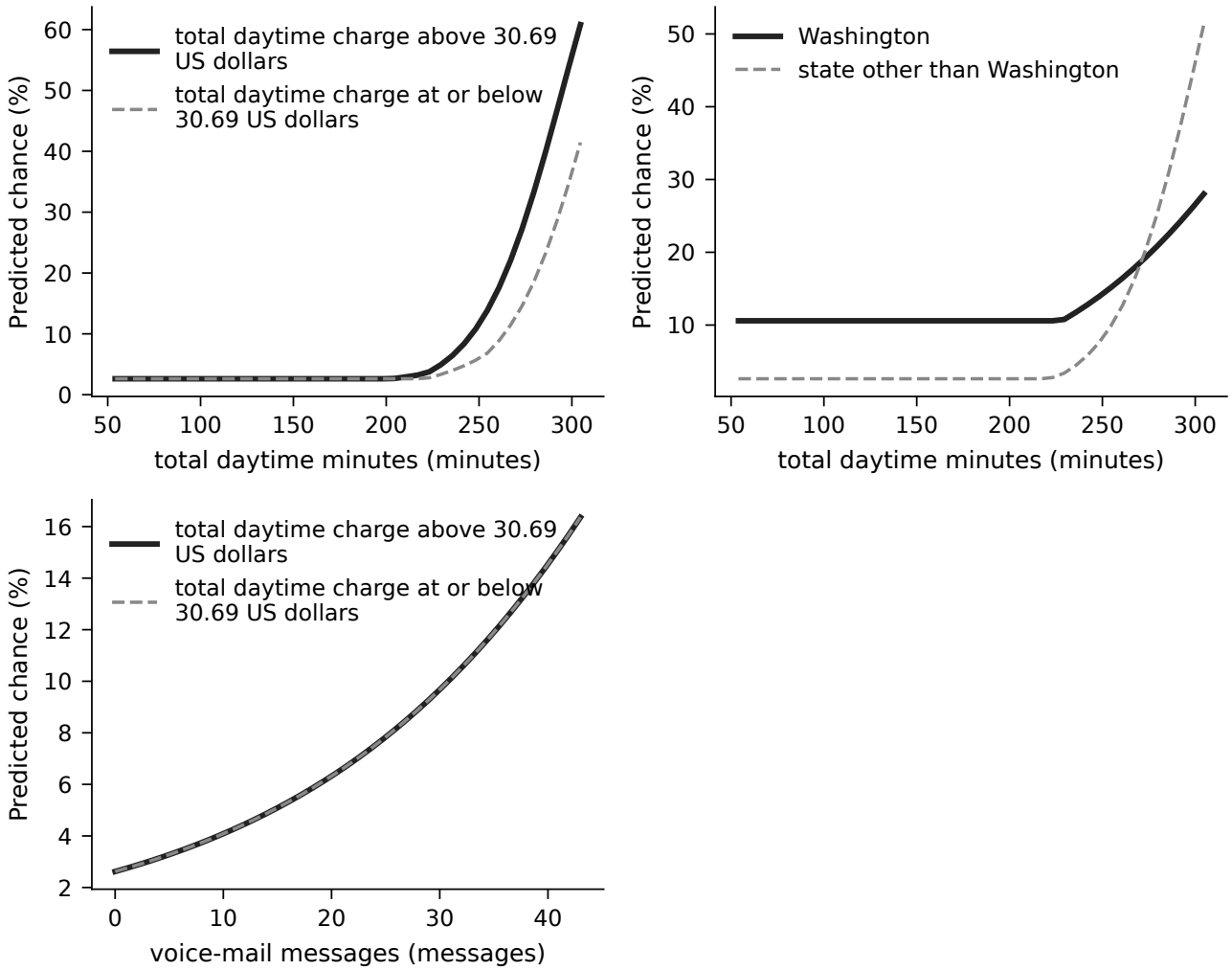


Figure 31: Heavy daytime use: How the effect differs between groups: one curve per group, for the typical case of each group.

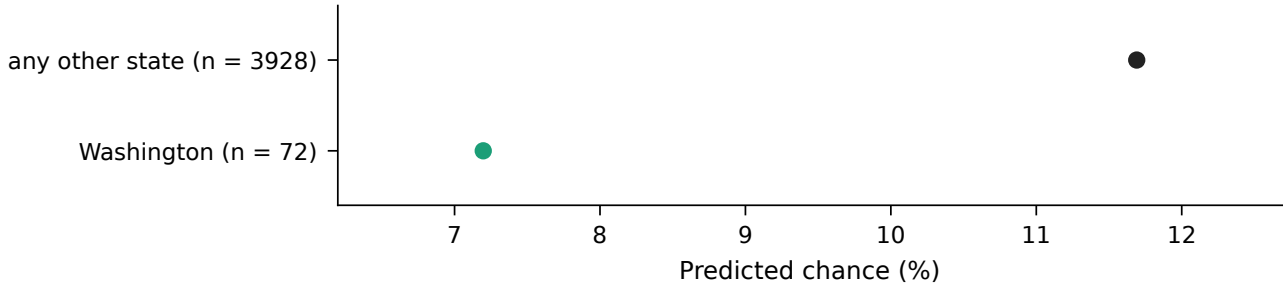


Figure 32: Heavy daytime use: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

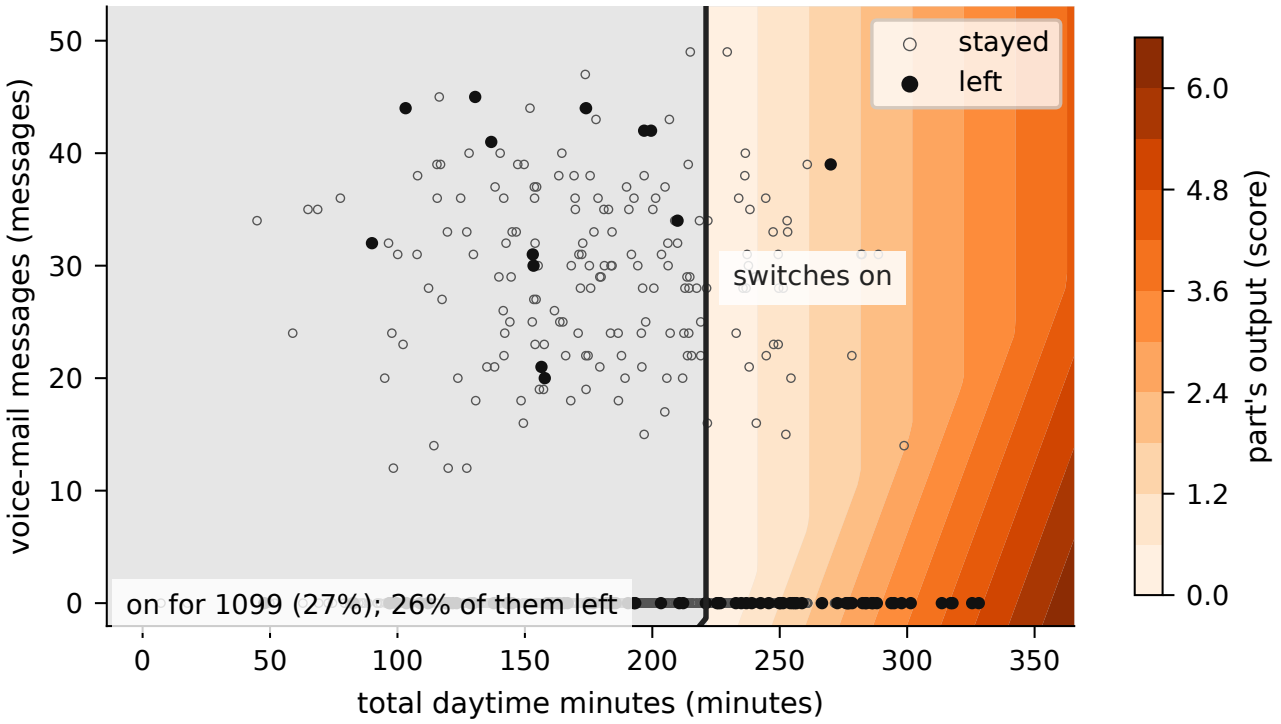


Figure 33: Heavy daytime use: The part's own output over its two numeric inputs, other inputs at the typical case: colour where it is on, flat grey where it is switched off, the black line where it switches on; training customers as hollow (no event) and filled (event) points, so the part can be seen catching the events (the predicted-chance overlay was harder to read). The switch-on line is traced on this map only; the boundary claim (C113) did not pass its check on the data, so no rule is stated.

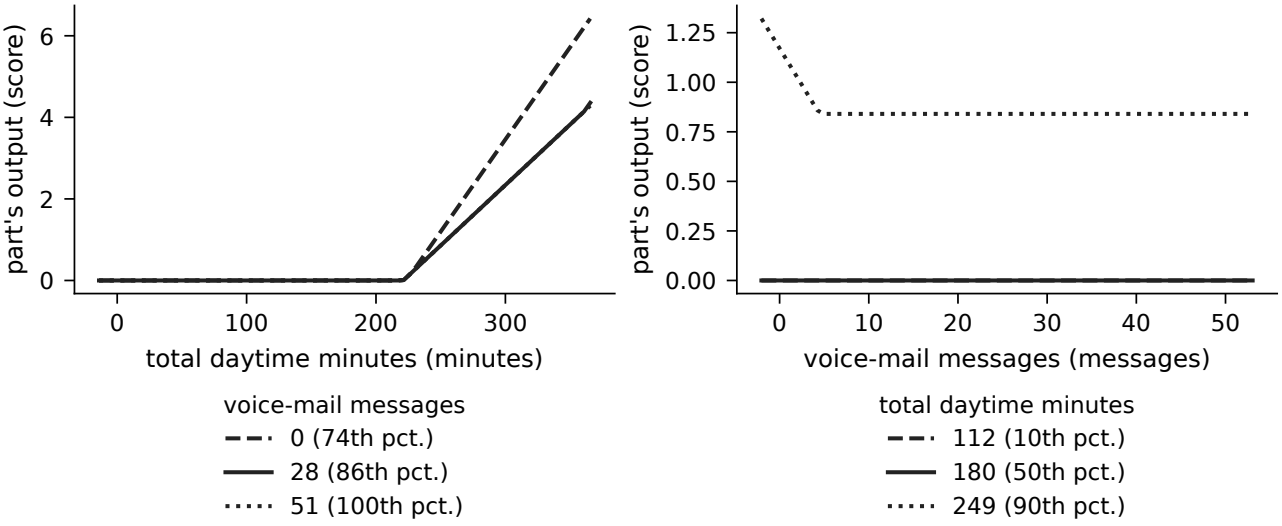


Figure 34: Heavy daytime use: Slices of the same map: the part's own output against each input at three values of the other, so the tilted boundary shows as a shifting hinge. The switch-on line is traced on this map only; the boundary claim (C113) did not pass its check on the data, so no rule is stated.

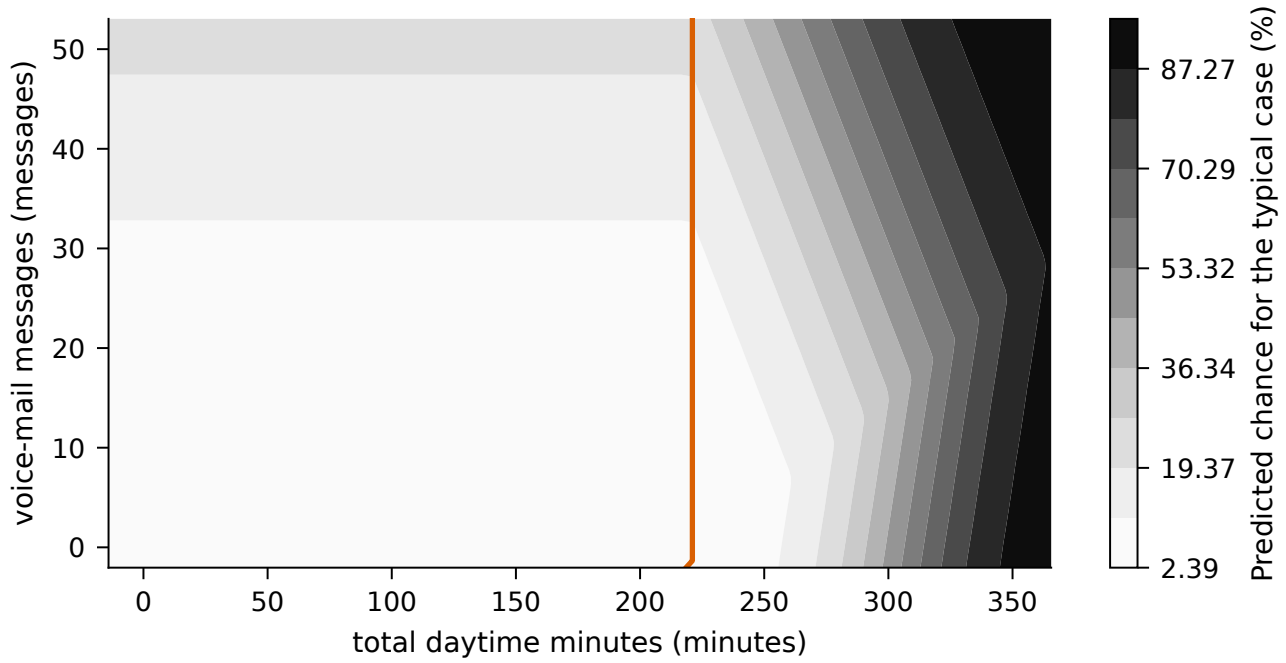


Figure 35: Heavy daytime use: The same map on the final scale: the predicted value for the typical case, with the switch-on line (orange). The switch-on line is traced on this map only; the boundary claim (C113) did not pass its check on the data, so no rule is stated.

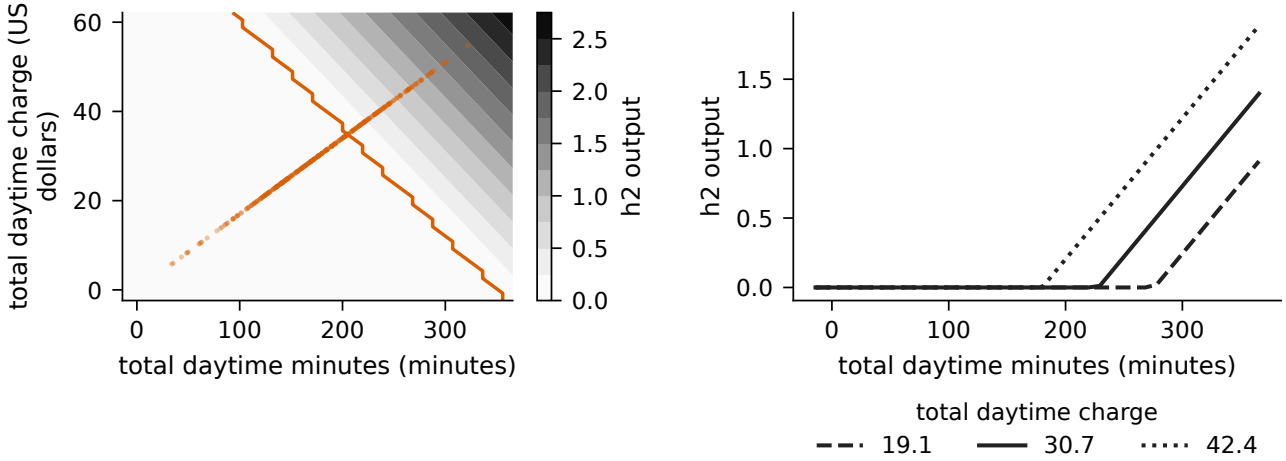


Figure 36: Heavy daytime use: This part's own output, unit h2 (the output of 'Heavy daytime use'), on its own scale (its activation), with every other input at the typical case; it is active for 896 of 4000. Left: its output over its two main inputs (orange line: where it switches on; dots: the data); right: slices at three values of the second input.

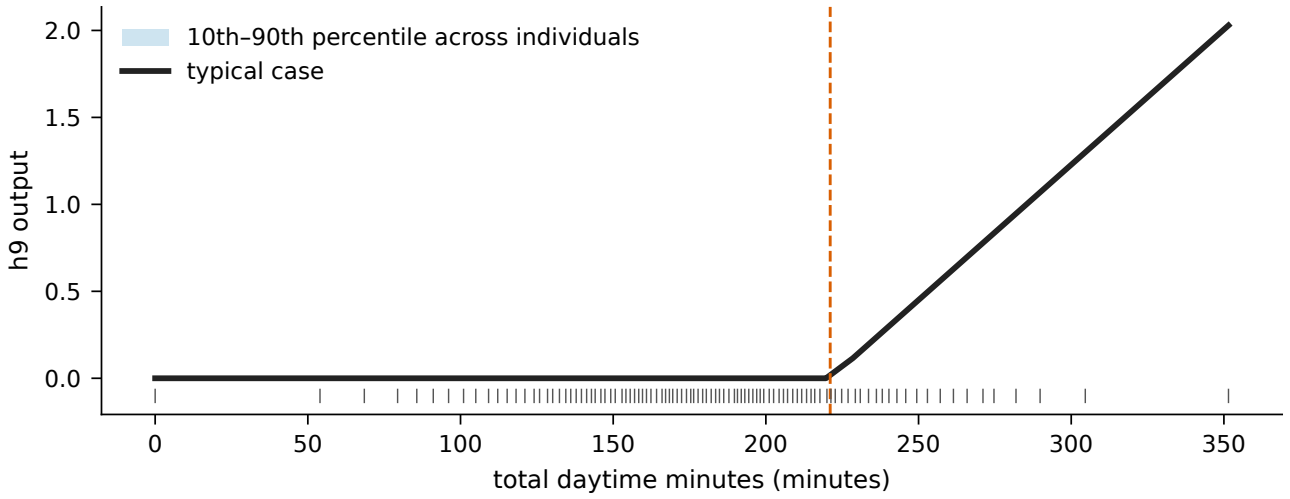


Figure 37: Heavy daytime use: This part's own output, unit h9 (the output of 'Heavy daytime use'), on its own scale (its activation), with every other input at the typical case; it is active for 870 of 4000. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

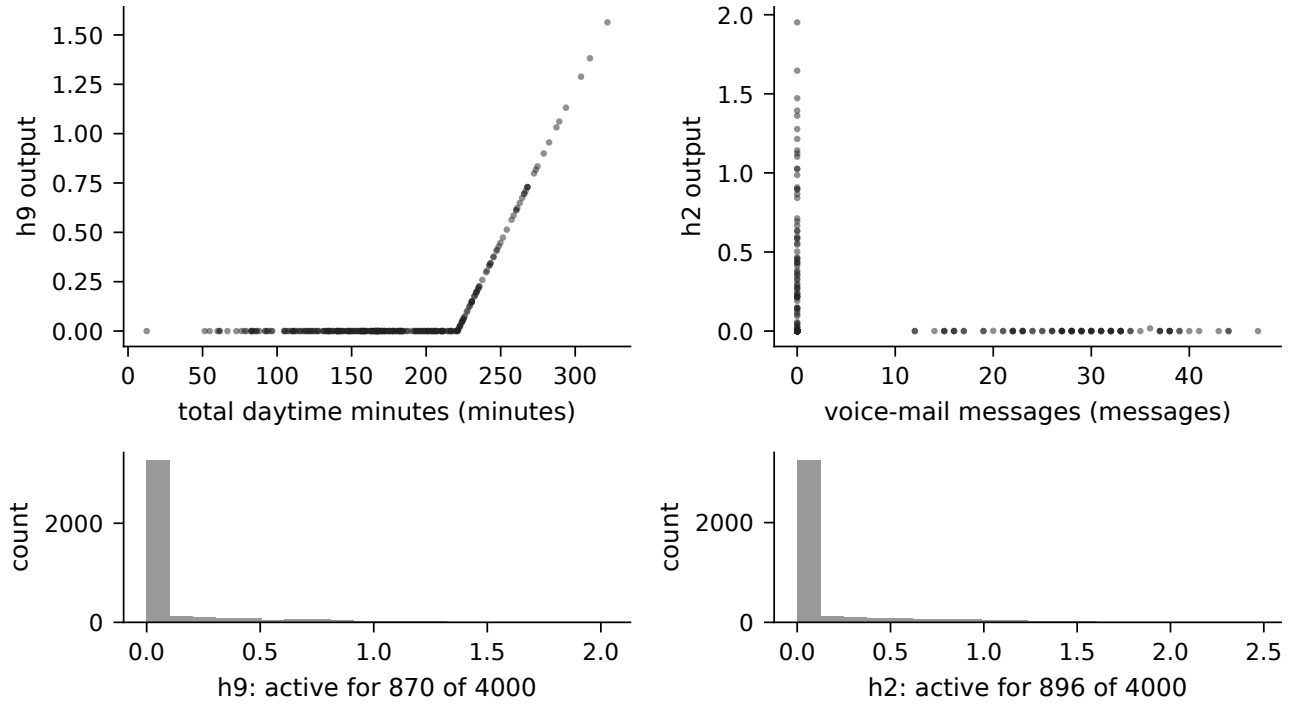


Figure 38: Heavy daytime use: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.6.

1.2.7 State units

Part of: The whole model (1.2.1).
Figure 39 shows what feeds this part; Figure 40 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 39: State units: what feeds it, one level down.

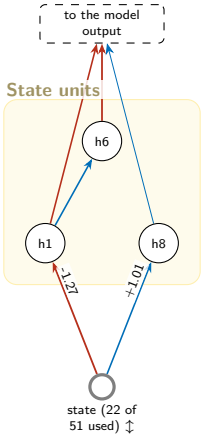


Figure 40: State units: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part reads only the state and raises the chance of leaving in Rhode Island, and a little in Indiana. [C100, C101]

Evidence. Through it, a typical customer's chance goes from about 3% to about 22% in Rhode Island, and up by less than a point in Indiana. [C100, C101] Its formula reproduces the part exactly on all 4000 training customers. [C102] Evidence plots: Figure 41, Figure 42.

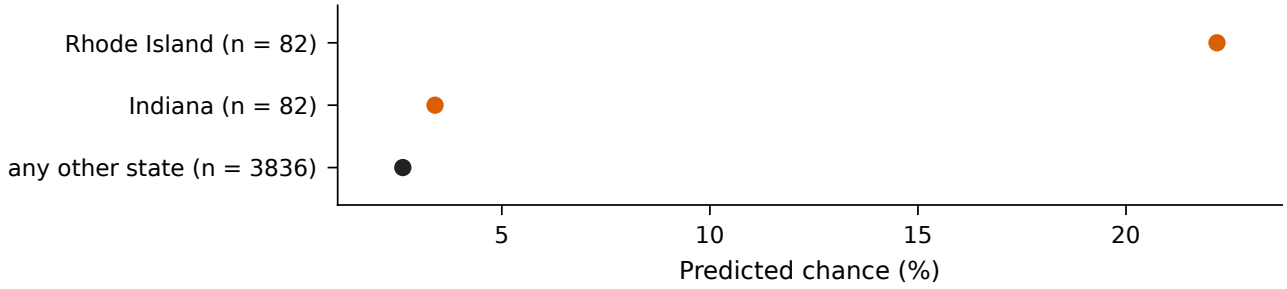


Figure 41: State units: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

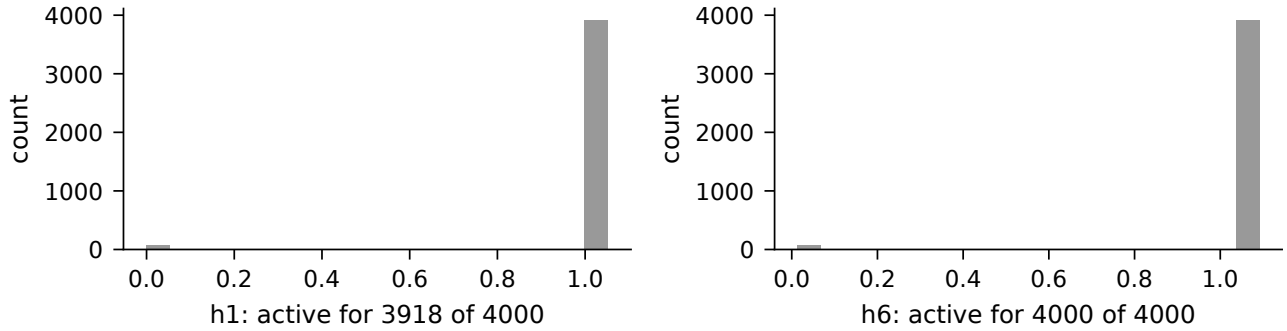


Figure 42: State units: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.7.

1.2.8 Evening unit

Part of: The whole model (1.2.1).
Figure 43 shows what feeds this part; Figure 44 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 43: Evening unit: what feeds it, one level down.

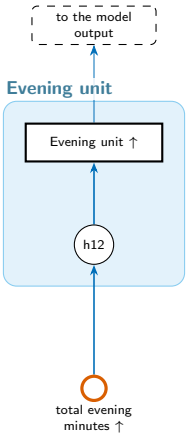


Figure 44: Evening unit: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part reads evening minutes and barely changes the chance of leaving: nothing until about 217 minutes, then steadily. [C103, C104]

Evidence. Through it, evening minutes change nothing until about 217 minutes, then raise the chance steadily, but a typical customer's chance stays at about 3% from 138 to 266 minutes. [C103] Its formula reproduces the part exactly on all 4000 training customers. [C105]
Evidence plots: Figure 45, Figure 46.

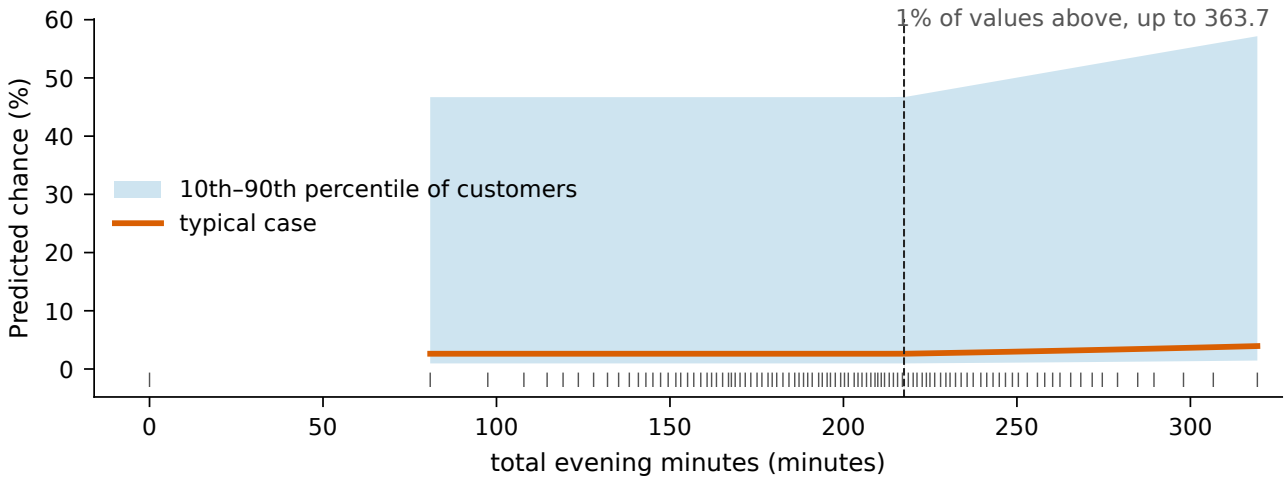


Figure 45: Evening unit: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

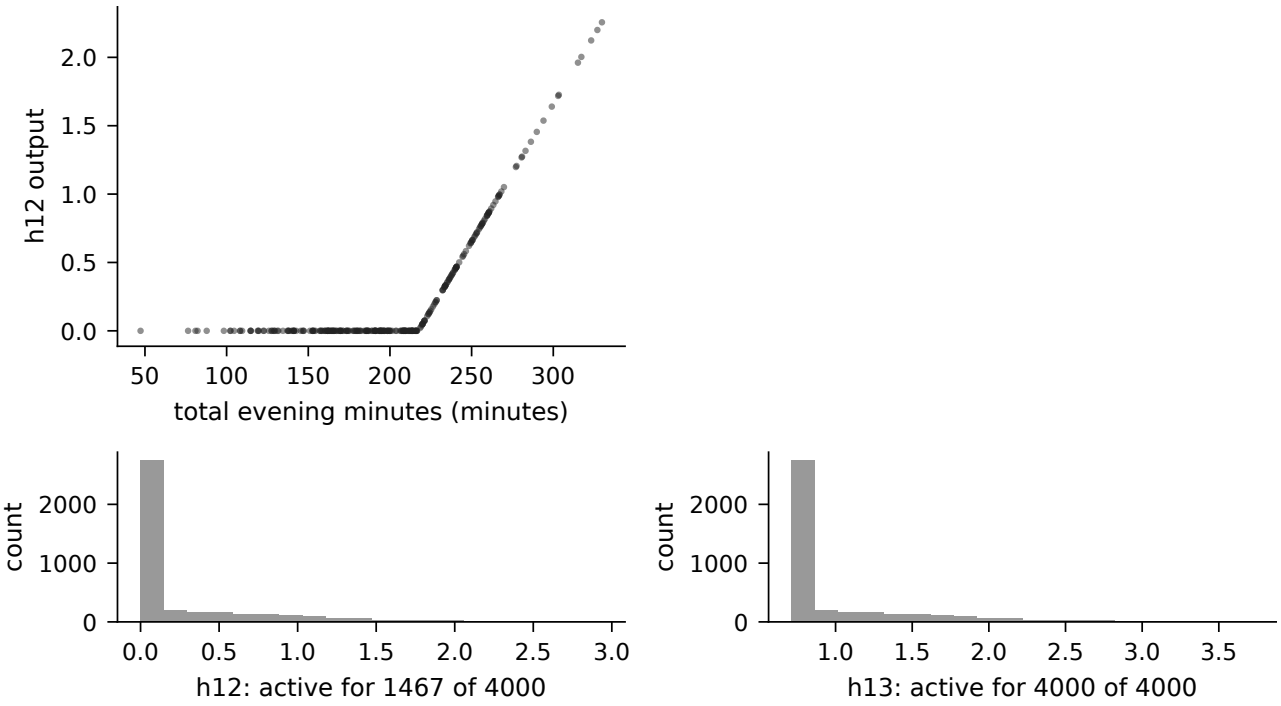


Figure 46: Evening unit: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.8.

1.2.9 Small voice-mail unit

Part of: The whole model (1.2.1).
Figure 47 shows what feeds this part; Figure 48 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 47: Small voice-mail unit: what feeds it, one level down.

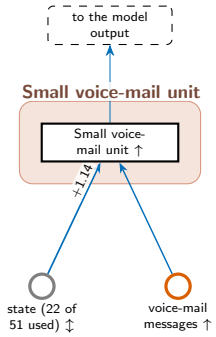


Figure 48: Small voice-mail unit: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This small part reads voice-mail messages and being in Louisiana, and barely changes the predictions. [C111]

Evidence. Through it, a typical customer's chance stays at about 3% in Louisiana as elsewhere, and voice-mail messages barely change it (about 3% at 0 and at 32 messages). [C106, C107] Switching it off changes the predictions by at most one point. [C111] Its formula reproduces the part exactly on all 4000 training customers. [C112]

Evidence plots: Figure 49, Figure 50, Figure 51, Figure 52.

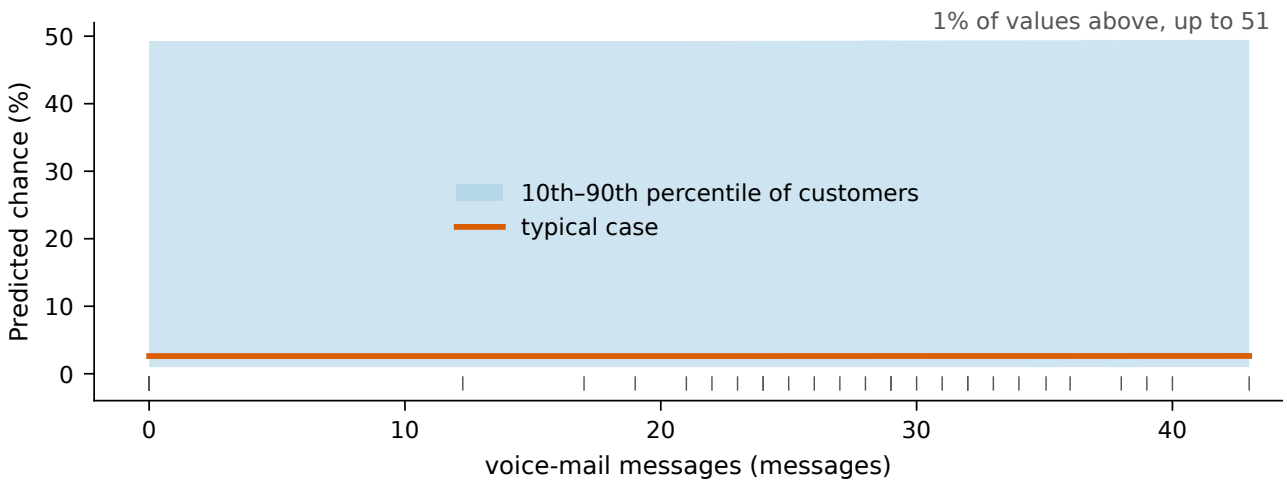


Figure 49: Small voice-mail unit: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

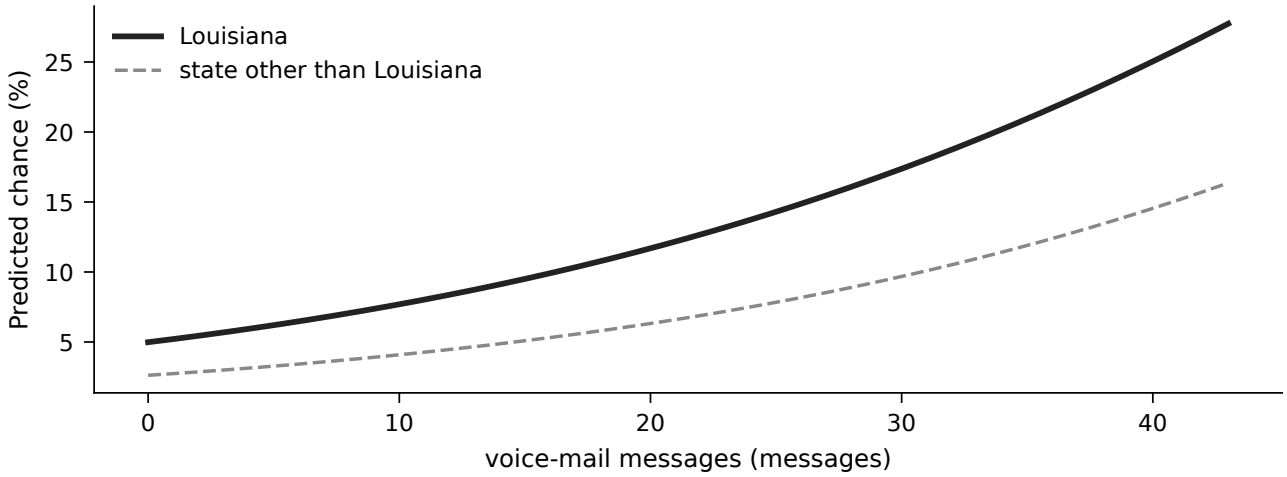


Figure 50: Small voice-mail unit: How the effect differs between groups: one curve per group, for the typical case of each group.

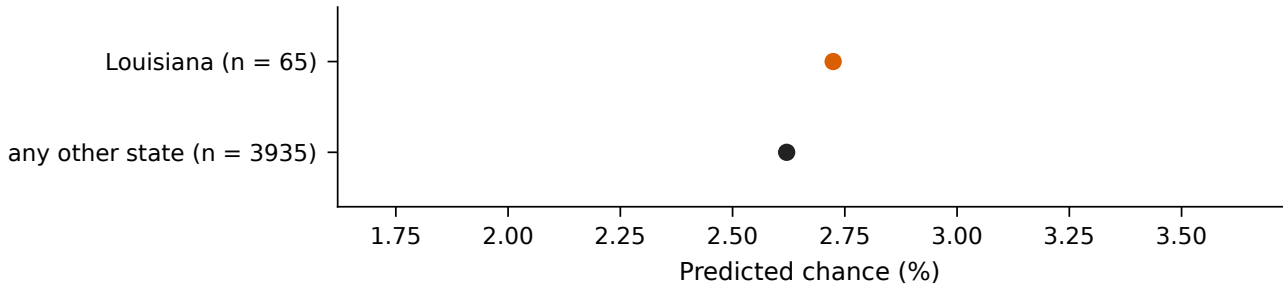


Figure 51: Small voice-mail unit: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

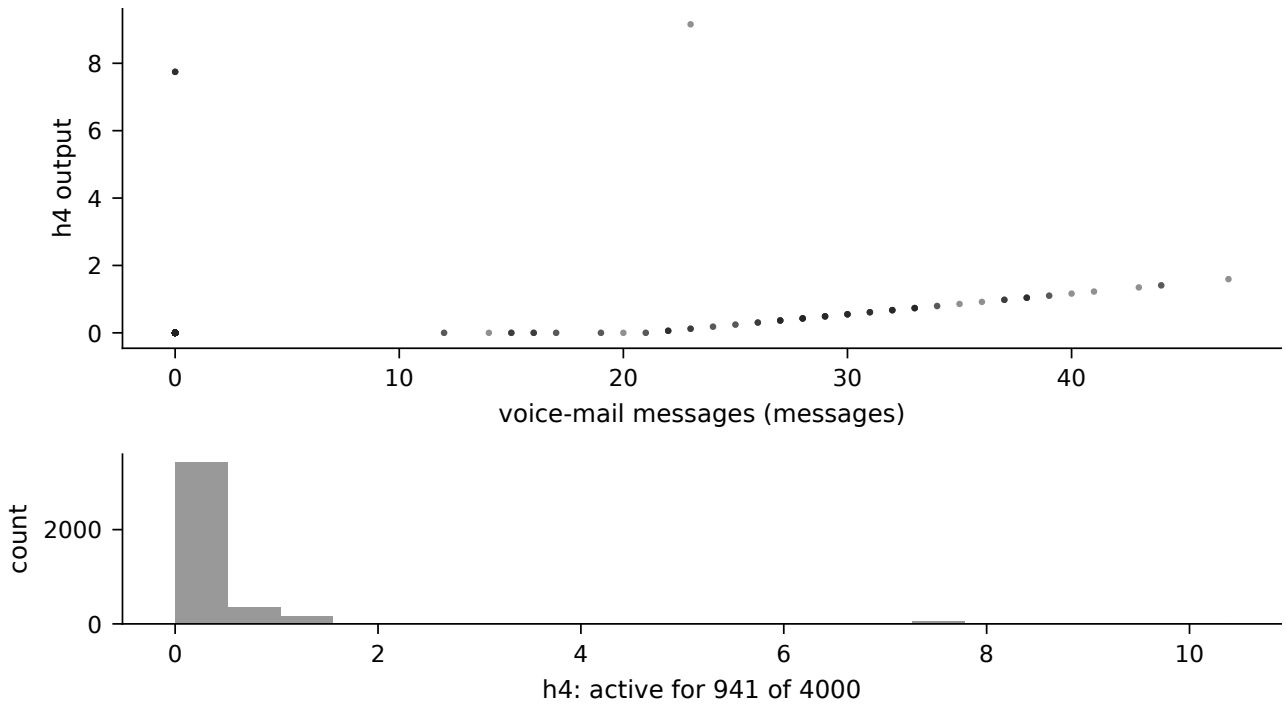


Figure 52: Small voice-mail unit: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.9.

1.3 Not supported by the model

- C109: threshold claim about number of voice-mail messages in part ann_84: not supported by the model (the fold changes the curve by only 0.007 log-odds compared with a straight line (needed 0.1); the effect past the fold is too small or not consistent across patients).
- C110: subgroup claim about state and number of voice-mail messages in part ann_84: not supported by the model (its effect is negligible).
- C113: boundary claim about total daytime minutes and number of voice-mail messages in part ann_82: not supported by the model (the boundary rule matches 0.0 of training rows (needs 0.99)).

Technical detail: churn100-s4-c2

The record behind each part: the original hypothesis, the closed form, and every claim with its estimate (log-odds and odds ratio), range across patients, n and test check.

T.1 The whole model

Annotation ann_87 (churn_logit). Reads: state, account length, area code, international calling plan, voice-mail plan, number of voice-mail messages, total daytime minutes, total daytime calls, total daytime charge, total evening minutes, total evening calls, total evening charge, total night-time minutes, total night-time calls, total night-time charge, total international minutes, total international calls, total international charge, number of calls to customer service.

Original hypothesis (unchanged). Root: logit P(churn) = -1.30 + linear_backbone (identity_lin) + service_calls (identity_svc) + 1.53*3118 + 1.91*10131 (day_usage_hinges) + 0.20*15524 (evening_hinge) - 1.01*1295 - 1.21*7872 + 0.04*9954 (state_offsets) + 0.005*6089, purely additive in the logit; the sigmoid is the only coupling. 37 inputs are unconnected (international charge and calls, night minutes/charge/calls, evening calls, area codes 408/510 and 29 state indicators) and have zero effect. Train AUC 0.8908, accuracy 0.888. Performance sources (train): all children replaced by least-squares linear fits gives AUC 0.8333 (an L2 logistic regression refit on all 71 inputs gets 0.8433); restoring day_usage_hinges alone gives 0.8707, service_calls alone 0.8659, both 0.8907; evening, state and voicemail units restore nothing. So the gain over linear (+0.058) comes from the day-usage and service-call thresholds, which are complementary. Rare-category switches: state:11 -2.75 logit (73 rows, empirical -1.35); state:46 -1.94 (80 rows, -0.91); state:45 -0.94 (91, -0.85); state:47 +0.90 mean (72 rows, empirical +0.92; direct +1.23, but it switches off 10131 so heavy day users lose up to 2.75); state:31 +0.83 (90, +0.67); state:18 +0.41 (65 rows, empirical -0.48: opposite sign); state:39 -0.25 net (82, -0.93: direct -2.61 almost cancelled by the state units).

Closed form. Read through its parts (their formulas are in their own sections).

Split inputs. To annotate the parts, the explanation split these inputs into copies (a refactoring step that does not change what the model computes); the diagrams draw each input once: customer-service calls (2 copies); voice-mail messages (3 copies).

State: every level the model reads. Whole-model change in the predicted chance for a typical customer when only the state changes from the reference; n = training customers in that level.

Claim	From	To	n	Chance	OK
C002	West Virginia	Washington	72	3% to 11%	yes
C005	West Virginia	New Jersey	90	3% to 7%	yes
C008	West Virginia	Louisiana	65	3% to 5%	yes
C009	West Virginia	Connecticut	80	3% to 5%	yes
C013	West Virginia	Tennessee	68	3% to 5%	yes
C014	West Virginia	Massachusetts	86	3% to 5%	yes
C015	West Virginia	Oregon	84	3% to 4%	yes
C016	West Virginia	Indiana	82	3% to 4%	yes
C017	West Virginia	Kentucky	81	3% to 4%	yes
C019	West Virginia	South Dakota	66	3% to 4%	yes
C020	West Virginia	Oklahoma	79	3% to 4%	yes
C024	West Virginia	North Dakota	68	3% to 3%	yes
C025	West Virginia	state other than Georgia, Hawaii, Indiana, Kentucky, Loui...	2286	3% to 3%	yes
C027	West Virginia	Arizona	73	3% to 3%	yes
C030	West Virginia	Arkansas	78	3% to 3%	yes
C033	West Virginia	Rhode Island	82	3% to 3%	yes
C034	West Virginia	Florida	78	3% to 3%	yes
C023	West Virginia	Pennsylvania	56	3% to 2%	yes
C022	West Virginia	Georgia	60	3% to 2%	yes
C018	West Virginia	Virginia	91	3% to 1%	yes
C012	West Virginia	Vermont	80	3% to <1%	yes
C007	West Virginia	Hawaii	73	3% to <1%	yes

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C001	direction: international calling plan: no international plan to an international plan (whole model) *shape: categorical; highest to lowest: an international plan, no international plan*	3% to 31%	2.82	16.7	2.82 to 2.82	3620, 380	agrees; AUC gain 0.060 [0.032, 0.090]	yes
C002	direction: state: West Virginia to Washington (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 11%	1.27	3.58	0.64 to 1.48	122, 72	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C003	direction: voice-mail messages: 0 messages to 32 messages (whole model) *shape: steady; 72% of patients same; fallback; risk saturates near 0%*	3% to 11%	1.21	3.35	0.39 to 1.47	4000	agrees; AUC gain 0.017 [0.005, 0.031]	yes
C004	direction: total daytime minutes: 112 minutes to 249 minutes (whole model) *shape: hinge at 223 minutes; 50% of patients same; fallback; risk saturates near 0%; matches the fold at 221 minutes*	3% to 8%	1.13	3.10	0.83 to 1.79	4000	agrees; AUC gain 0.036 [0.015, 0.059]	yes
C005	direction: state: West Virginia to New Jersey (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 7%	1.09	2.97	1.09 to 1.09	122, 90	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C006	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (whole model) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	3% to <1%	-2.54	0.08	-2.54 to -2.54	2939, 1061	agrees; AUC gain 0.033 [0.014, 0.053]	yes
C007	direction: state: West Virginia to Hawaii (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to <1%	-2.50	0.08	-2.50 to -2.50	122, 73	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C008	direction: state: West Virginia to Louisiana (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 5%	0.67	1.95	0.67 to 0.67	122, 65	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C009	direction: state: West Virginia to Connecticut (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 5%	0.66	1.94	0.66 to 0.66	122, 80	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C010	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 4%	0.88	2.42	0.88 to 0.88	4000	agrees; AUC gain 0.013 [0.004, 0.022]	yes
C011	direction: total international minutes: 7 minutes to 14 minutes (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 4%	0.85	2.35	0.85 to 0.85	4000	unclear; AUC gain 0.0008 [-0.007, 0.008]	yes
C012	direction: state: West Virginia to Vermont (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to <1%	-1.68	0.19	-1.68 to -1.68	122, 80	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C013	direction: state: West Virginia to Tennessee (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 5%	0.60	1.82	0.60 to 0.60	122, 68	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C014	direction: state: West Virginia to Massachusetts (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 5%	0.57	1.77	0.57 to 0.57	122, 86	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C015	direction: state: West Virginia to Oregon (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 4%	0.53	1.70	0.53 to 0.53	122, 84	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C016	direction: state: West Virginia to Indiana (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 4%	0.52	1.69	0.52 to 0.52	122, 82	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C017	direction: state: West Virginia to Kentucky (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 4%	0.47	1.61	0.47 to 0.47	122, 81	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C018	direction: state: West Virginia to Virginia (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 1%	-0.69	0.50	-0.69 to -0.69	122, 91	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C019	direction: state: West Virginia to South Dakota (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 4%	0.32	1.37	0.32 to 0.32	122, 66	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C020	direction: state: West Virginia to Oklahoma (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 4%	0.31	1.36	0.31 to 0.31	122, 79	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C021	direction: total daytime calls: 75 calls to 125 calls (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 3%	0.33	1.40	0.33 to 0.33	4000	disagrees; AUC gain -0.002 [-0.005, 0.0008]	yes
C022	direction: state: West Virginia to Georgia (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 2%	-0.40	0.67	-0.40 to -0.40	122, 60	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C023	direction: state: West Virginia to Pennsylvania (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 2%	-0.37	0.69	-0.37 to -0.37	122, 56	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C024	direction: state: West Virginia to North Dakota (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 3%	0.26	1.30	0.26 to 0.26	122, 68	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C025	direction: state: West Virginia to state other than Georgia, Hawaii, Indiana, Kentucky, Louisiana, Massachusetts, Arkansas, North Dakota, Arizona, New Jersey, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Dakota, Tennessee, Virginia, Vermont, Washington, West Virginia, Connecticut, Florida (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 3%	0.26	1.29	0.26 to 0.26	122, 2286	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C026	direction: account length: 48 to 152 (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 3%	0.28	1.32	0.28 to 0.28	4000	disagrees; AUC gain -0.001 [-0.004, 0.0006]	yes
C027	direction: state: West Virginia to Arizona (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 3%	0.22	1.25	0.22 to 0.22	122, 73	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C028	direction: total evening minutes: 138 minutes to 266 minutes (whole model) *shape: hinge at 218 minutes; 100% of patients same; verified; risk saturates near 0%; matches the fold at 217 minutes*	3% to 3%	0.20	1.22	0.20 to 0.20	4000	unclear; AUC gain 0.003 [-8e-05, 0.005]	yes
C029	direction: customer-service calls: 0 calls to 3 calls (whole model) *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	3% to 3%	0.19	1.21	0.19 to 0.19	4000	agrees; AUC gain 0.111 [0.070, 0.152]	yes
C030	direction: state: West Virginia to Arkansas (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 3%	0.14	1.15	0.14 to 0.14	122, 78	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C031	direction: area code: area code 415 to area code other than area code 415 (whole model) *shape: categorical; highest to lowest: area code 415, area code other than area code 415*	3% to 3%	-0.01	0.99	-0.01 to -0.01	2016, 1984	disagrees; AUC gain -0.0002 [-0.0004, 5e-05]	yes
C032	direction: total daytime charge: 19.07 US dollars to 42.40 US dollars (whole model) *shape: hinge at 42.20 US dollars; 77% of patients same; fallback; risk saturates near 0%; matches the fold at 39.34 US dollars*	3% to 3%	0.24	1.27	0.00 to 0.94	4000	unclear; AUC gain 0.005 [-0.0004, 0.012]	yes
C033	direction: state: West Virginia to Rhode Island (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 3%	0.004	1.00	0.004 to 0.004	122, 82	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C034	direction: state: West Virginia to Florida (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 3%	-0.0006	1.00	-0.0006 to -0.0006	122, 78	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C035	magnitude: international calling plan: no international plan to an international plan, typical patient *shape: categorical; highest to lowest: an international plan, no international plan*	3% to 31%	2.82	16.7		3620, 380	agrees; AUC gain 0.060 [0.032, 0.090]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C036	magnitude: state: West Virginia to Washington, typical patient *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 11%	1.48	4.40		122, 72	disagrees; AUC gain -0.012 [-0.026, 0.0005]	yes
C037	magnitude: voice-mail messages: 0 messages to 32 messages, typical patient *shape: steady; 72% of patients same; fallback; risk saturates near 0%*	3% to 11%	1.47	4.37		4000	agrees; AUC gain 0.017 [0.005, 0.031]	yes
C038	magnitude: total daytime minutes: 112 minutes to 249 minutes, typical patient *shape: hinge at 223 minutes; 50% of patients same; fallback; risk saturates near 0%; matches the fold at 221 minutes*	3% to 8%	1.15	3.17		4000	agrees; AUC gain 0.036 [0.015, 0.059]	yes
C039	magnitude: voice-mail plan: no voice-mail plan to a voice-mail plan, typical patient *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	3% to <1%	-2.54	0.08		2939, 1061	agrees; AUC gain 0.033 [0.014, 0.053]	yes
C040	null: model ignores total evening calls (exact rule)		0.00			4000	train only	yes
C041	null: model ignores total international calls (exact rule)		0.00			4000	train only	yes
C042	null: model ignores total international charge (exact rule)		0.00			4000	train only	yes
C043	null: model ignores total night-time calls (exact rule)		0.00			4000	train only	yes
C044	null: model ignores total night-time charge (exact rule)		0.00			4000	train only	yes
C045	null: model ignores total night-time minutes (exact rule)		0.00			4000	train only	yes
C046	structure: made of linear_backbone, service_calls, day_usage_hinges, evening_hinge, vmail_state18_unit, state_offsets; added					4000	train only	yes

T.2 Fixed-weight inputs

Annotation ann_79 (linear_backbone). Reads: state, account length, area code, international calling plan, voice-mail plan, number of voice-mail messages, total daytime calls, total evening charge, total international minutes.

Original hypothesis (unchanged). Direct input-to-output edges collected through identity_lin (an identity node). Exactly linear in 29 inputs: voicemail plan -1.12 per SD (-2.54 logit for having a plan, mostly offset for heavy message users by voicemail messages +0.62 per SD), international plan +0.83 (+2.82 logit for having one), evening charge +0.35, international minutes +0.34, day calls +0.13, account length +0.11, and state offsets: state:11 -0.37 (-2.75 logit for its 73 customers), state:39 -0.37 (-2.61, but the state_offsets units give back +2.36), state:46 -0.27 (-1.94), state:47 +0.16 (+1.23), state:31 +0.12 (+0.83), state:45 -0.14 (-0.94), then smaller states and area code 415 (+0.01). Day minutes, day charge and customer-service calls have NO direct edge: they act only through hinges. Evidence: closed form exact (identity); train sensitivities equal the weights (voicemail plan 1.12, voicemail messages 0.62, international plan 0.83).

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 1.33227e-15).

$$
\begin{aligned}
y_0 = {} & 0.00719x_{areaCode:415} + 0.106x_{accountLength} + 0.0569x_{state:6} \\
& + 0.163x_{state:47_a} + 0.133x_{totalDayCalls} + 0.337x_{totalIntlMinutes} \\
& + 0.044x_{state:42} + 0.007x_{state:36} + 0.00781x_{state:41} \\
& + 0.621x_{numberVmailMessages_a} + 0.0397x_{state:37} + 0.35x_{totalEveCharge} \\
& + 0.0453x_{state:19} + 0.047x_{state:18_a} + 0.826x_{internationalPlan} \\
& + 0.000897x_{state:28} + 0.0307x_{state:17} + 0.124x_{state:31} - 0.37x_{state:39_a} \\
& - 0.0355x_{state:9} - 0.0158x_{state:2} - 0.00422x_{state:3} - 0.271x_{state:46} \\
& - 1.12x_{voiceMailPlan} - 0.141x_{state:45} - 0.044x_{state:49} - 0.0737x_{state:38} \\
& - 0.0797x_{state:10} - 0.368x_{state:11}
\end{aligned}
$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C047	direction: international calling plan: no international plan to an international plan (through this part) *shape: categorical; highest to lowest: an international plan, no international plan*	3% to 31%	2.82	16.7	2.82 to 2.82	3620, 380	agrees; AUC gain 0.060 [0.032, 0.090]	yes
C048	direction: state: West Virginia to Washington (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 11%	1.48	4.40	1.48 to 1.48	122, 72	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C049	direction: voice-mail messages: 0 messages to 32 messages (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	3% to 10%	1.47	4.35	1.47 to 1.47	4000	agrees; AUC gain 0.016 [0.003, 0.029]	yes
C050	direction: state: West Virginia to New Jersey (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 7%	1.09	2.97	1.09 to 1.09	122, 90	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C051	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	3% to <1%	-2.54	0.08	-2.54 to -2.54	2939, 1061	agrees; AUC gain 0.033 [0.014, 0.053]	yes
C052	direction: state: West Virginia to Hawaii (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to <1%	-2.50	0.08	-2.50 to -2.50	122, 73	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C053	direction: state: West Virginia to Rhode Island (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to <1%	-2.36	0.09	-2.36 to -2.36	122, 82	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C054	direction: state: West Virginia to Connecticut (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 5%	0.66	1.94	0.66 to 0.66	122, 80	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C055	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 4%	0.88	2.42	0.88 to 0.88	4000	agrees; AUC gain 0.013 [0.004, 0.022]	yes
C056	direction: total international minutes: 7 minutes to 14 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 4%	0.85	2.35	0.85 to 0.85	4000	unclear; AUC gain 0.0008 [-0.007, 0.008]	yes
C057	direction: state: West Virginia to Louisiana (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 5%	0.63	1.87	0.63 to 0.63	122, 65	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C058	direction: state: West Virginia to Vermont (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to <1%	-1.68	0.19	-1.68 to -1.68	122, 80	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C059	direction: state: West Virginia to Tennessee (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 5%	0.60	1.82	0.60 to 0.60	122, 68	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C060	direction: state: West Virginia to Massachusetts (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 5%	0.57	1.77	0.57 to 0.57	122, 86	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C061	direction: state: West Virginia to Oregon (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 4%	0.53	1.70	0.53 to 0.53	122, 84	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C062	direction: state: West Virginia to Kentucky (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 4%	0.47	1.61	0.47 to 0.47	122, 81	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C063	direction: state: West Virginia to Virginia (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 1%	-0.69	0.50	-0.69 to -0.69	122, 91	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C064	direction: state: West Virginia to South Dakota (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 4%	0.32	1.37	0.32 to 0.32	122, 66	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C065	direction: state: West Virginia to Oklahoma (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts ...*	3% to 4%	0.31	1.36	0.31 to 0.31	122, 79	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C066	direction: total daytime calls: 75 calls to 125 calls (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 3%	0.33	1.40	0.33 to 0.33	4000	disagrees; AUC gain -0.002 [-0.005, 0.0008]	yes
C067	direction: state: West Virginia to Georgia (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts ...*	3% to 2%	-0.40	0.67	-0.40 to -0.40	122, 60	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C068	direction: state: West Virginia to Pennsylvania (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts ...*	3% to 2%	-0.37	0.69	-0.37 to -0.37	122, 56	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C069	direction: state: West Virginia to North Dakota (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts ...*	3% to 3%	0.26	1.30	0.26 to 0.26	122, 68	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C070	direction: state: West Virginia to state other than Georgia, Hawaii, Kentucky, Louisiana, Massachusetts, Arkansas, North Dakota, Arizona, New Jersey, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Dakota, Tennessee, Virginia, Vermont, Washington, West Virginia, Connecticut, Florida (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts ...*	3% to 3%	0.26	1.29	0.26 to 0.26	122, 2368	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C071	direction: account length: 48 to 152 (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 3%	0.28	1.32	0.28 to 0.28	4000	disagrees; AUC gain -0.001 [-0.004, 0.0006]	yes
C072	direction: state: West Virginia to Arizona (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts ...*	3% to 3%	0.22	1.25	0.22 to 0.22	122, 73	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C073	direction: state: West Virginia to Arkansas (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts ...*	3% to 3%	0.14	1.15	0.14 to 0.14	122, 78	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C074	direction: area code: area code 415 to area code other than area code 415 (through this part) *shape: categorical; highest to lowest: area code 415, area code other than area code 415*	3% to 3%	-0.01	0.99	-0.01 to -0.01	2016, 1984	disagrees; AUC gain -0.0002 [-0.0004, 5e-05]	yes
C075	direction: state: West Virginia to Florida (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts ...*	3% to 3%	-0.0006	1.00	-0.0006 to -0.0006	122, 78	disagrees; AUC gain -0.008 [-0.022, 0.005]	yes
C076	structure: its formula reproduces it on 4000 training patients (largest difference 1.33227e-15)		1e-15			4000	train only	yes

T.3 Service calls

Annotation ann_86 (service_calls). Reads: number of calls to customer service.

Original hypothesis (unchanged). Composition of csc_moderate_chain and csc_high_hinge: their exits are summed by identity_svc = 1.66*5268 + 1.76*9436 + 1.54*6713, which enters the logit with weight 1; they read the same input (two split copies of customer-service calls). Hypothesis: all of the model's service-call effect is a threshold: nothing for 0-2 calls, +0.19 logit at 3 calls, then about +2.4 logit per extra call (+2.07 at 4, +4.44 at 5, +6.81 at 6, +9.18 at 7). Evidence (train): churn by calls 0-7 is 0.115/0.110/0.110/0.106/0.437/0.571/0.679/0.545 against model 0.118/0.108/0.105/0.115/0.327/0.749/0.956/0.986: the threshold is right, but the model under-predicts at 4 calls and is over-confident from 5 calls (116 customers). Mean-ablating both children gives AUC 0.7976, linear replacement 0.8711; restoring service_calls alone onto the all-linear model gives 0.8659 (all-linear 0.8333).

Closed form. Read through its parts (their formulas are in their own sections).

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C077	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	3% to 3%	0.19	1.21	0.19 to 0.19	4000	agrees; AUC gain 0.111 [0.070, 0.152]	yes
C078	threshold: customer-service calls: fold at about 3 calls (starts) *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	3% to 18% (typical patient with a fold)	2.07	7.90	2.07 to 2.07	4000, 848	agrees; AUC gain 0.111 [0.070, 0.152]	yes
C079	structure: made of csc_moderate_chain, csc_high_hinge; added					4000	train only	yes

T.4 Calls from 4 upwards

Annotation ann_81 (csc_high_hinge). Reads: number of calls to customer service.

Original hypothesis (unchanged). 9436 = relu(-0.58 + 0.46 csc), 6713 = relu(-0.13 + 1.43*9436); logit weights +1.76 and +1.54 (via identity_svc). Hypothesis: a second hinge on customer-service calls, off for 0-3 calls and switching on from 4 calls, adding about +1.4 logit per extra call (+0.91 at 4, +2.32 at 5, +3.72 at 6). Active on exactly the 8% of customers with 4+ calls (318 rows; empirical churn jumps from 0.11 at 3 calls to 0.44 at 4). Evidence: line plot correlation 0.66; mean ablation AUC 0.8761; linear replacement 0.8857 (R2 0.44/0.41); restored alone onto all-linear 0.8571.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 0.0).

$$
\begin{aligned}
y_0 &= \operatorname{relu}\big(-0.582 + 0.462 x_{numberCustomerServiceCalls_b}\big) \\
y_1 &= \operatorname{relu}\big(-0.127 + 1.43 \operatorname{relu}\big(-0.582 \\
&\qquad + 0.462 x_{numberCustomerServiceCalls_b}\big)\big)
\end{aligned}
$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C093	threshold: customer-service calls: fold at about 4 calls (starts) *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	3% to 6% (typical patient with a fold)	0.91	2.49	0.91 to 0.91	4000, 318	agrees; AUC gain 0.023 [0.012, 0.035]	yes
C094	subgroup: who this part acts on: 318 of 4000 patients	8% to 18% (typical patient this part acts on (risk without the part, then with it))	1.71	5.51	0.91 to 3.72	318, 3682	agrees; AUC gain 0.023 [0.012, 0.035]	yes
C095	structure: its formula reproduces it on 4000 training patients (largest difference 0.0)		0.00			4000	train only	yes

T.5 Calls from 3 upwards

Annotation ann_80 (csc_moderate_chain). Reads: number of calls to customer service.

Original hypothesis (unchanged). 12730 = relu(-0.03 + 0.80 csc), 12171 = relu(-0.44 + 0.84 csc + 0.72*12730), 5268 = relu(-0.56 + 0.25 csc + 0.36*12171); logit weight +1.66 (via identity_svc). Hypothesis: a convex hinge on customer-service calls that is zero for 0-2 calls, just switches on at 3 calls (mean logit +0.19) and then adds about +1.0 logit per extra call (+1.16 at 4, +2.12 at 5, +3.09 at 6). Evidence (train, n=4000): line plot of 5268 against calls rises from 0 to 3.6 (correlation 0.74); 5268 active on 21% of customers (all with 3+ calls). Mean ablation AUC 0.8733 (full 0.8908); linear replacement 0.8852 (R2 0.55); restored alone onto the all-linearised model 0.8565 (all-linear 0.8333).

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 4.44089e-16).

$$\begin{aligned} y_0 = \operatorname{relu}\big(&-0.557 + 0.358 \operatorname{relu}\big(-0.443 \\ &+ 0.719 \operatorname{relu}\big(-0.0259 + 0.801 x_{numberCustomerServiceCalls_acd}\big) \\ &+ 0.845 x_{numberCustomerServiceCalls_acd}\big) \\ &+ 0.254 x_{numberCustomerServiceCalls_acd}\big) \end{aligned}$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C096	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	3% to 3%	0.19	1.21	0.19 to 0.19	4000	agrees; AUC gain 0.028 [0.014, 0.042]	yes
C097	threshold: customer-service calls: fold at about 3 calls (starts) *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	3% to 8% (typical patient with a fold)	1.16	3.18	1.16 to 1.16	4000, 848	agrees; AUC gain 0.028 [0.014, 0.042]	yes
C098	subgroup: who this part acts on: 848 of 4000 patients	3% to 4% (typical patient this part acts on (risk without the part, then with it))	0.76	2.13	0.19 to 2.12	848, 3152	agrees; AUC gain 0.028 [0.014, 0.042]	yes
C099	structure: its formula reproduces it on 4000 training patients (largest difference 4.44089e-16)		4e-16			4000	train only	yes

T.6 Heavy daytime use

Annotation ann_82 (day_usage_hinges). Reads: state, number of voice-mail messages, total daytime minutes, total daytime charge.

Original hypothesis (unchanged). 3118 = relu(-0.86 + 0.55 dayMin + 0.38 dayCharge - 0.65 vmailMessages), logit weight +1.53; 10131 = relu(-0.72 + 0.84 dayMin - 0.67 state:47), logit weight +1.91. Day minutes and day charge are collinear (r = 1.0000), so 3118 is one hinge in day usage. Hypothesis: day usage has no effect below about 200 minutes, then raises churn risk steeply. 3118 switches on at about 209 day minutes for customers with no voicemail messages (its kink moves up about 25 minutes per 10 messages: 267 at 20, 292 at 30), so it is on for only 2% of message users; 10131 switches on at about 221 day minutes for everyone except state:47 (off for all 72 state:47 customers). Evidence (train): combined logit contribution 0 below 200 minutes, +0.62 at 220-240, +1.61 at 240-260, +2.59 at 260-280, +4.26 above 280; churn by band 0.06 (200-220), 0.14, 0.26, 0.46, 0.67 vs model 0.08, 0.15, 0.28, 0.43, 0.68 (well calibrated). Sensitivity: 3118 vmail messages 0.15, day minutes 0.12, day charge 0.09; 10131 day minutes 0.18, state:47 0.14. Mean ablation AUC 0.8107 (the largest single drop); linear replacement 0.8657 (R2 0.46/0.45); restored alone onto all-linear 0.8707 (all-linear 0.8333).

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 4.44089e-16).

$$
\begin{aligned}
y_0 = \mathrm{relu}\big(&-0.856 + 0.382 x_{totalDayCharge} + 0.547 x_{totalDayMinutes} \\
&- 0.65 x_{numberVmailMessages_b}\big) \\
y_1 = \mathrm{relu}\big(&-0.721 + 0.837 x_{totalDayMinutes} \\
&- 0.672 x_{state:47_b}\big)
\end{aligned}
$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C080	direction: voice-mail messages: 0 messages to 32 messages (through this part) *shape: hinge at 11 messages; 64% of patients same; fallback; one point with C089 (median fold 11 messages, typical patient 11 messages, 10th-90th percentile 4 messages to 27 messages, wide)*	12% to 6% (typical patient this part acts on)	-0.26	0.77	-1.08 to 0.00	4000	agrees; AUC gain 0.006 [0.003, 0.009]	yes
C081	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: hinge at 221 minutes; 50% of patients same; fallback; risk saturates near 0%; one point with C088 (median fold 221 minutes, typical patient 223 minutes, 10th-90th percentile 186 minutes to 221 minutes, wide)*	3% to 8%	1.13	3.10	0.83 to 1.79	4000	agrees; AUC gain 0.036 [0.015, 0.059]	yes
C082	direction: state: state other than Washington to Washington (through this part) *shape: categorical; highest to lowest: state other than Washington, Washington*	12% to 7% (typical patient this part acts on)	-0.21	0.81	-0.84 to 0.00	3928, 72	agrees; AUC gain 0.0004 [2e-05, 0.0008]	yes
C083	direction: total daytime charge: 19.07 US dollars to 42.40 US dollars (through this part) *shape: hinge at 39.34 US dollars; 77% of patients same; fallback; risk saturates near 0%; one point with C090 (median fold 39.34 US dollars, typical patient 42.20 US dollars, 10th-90th percentile 26.14 US dollars to 52.75 US dollars, wide)*	3% to 3%	0.24	1.27	0.00 to 0.94	4000	unclear; AUC gain 0.005 [-0.0004, 0.012]	yes
C084	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by total daytime charge: total daytime charge above 30.69 US dollars minus total daytime charge at or below 30.69 US dollars; stronger in total daytime charge above 30.69 US dollars *shape: accelerating; 74% of patients same; fallback; risk saturates near 0%*	total daytime charge above 30.69 US dollars: 3% to 11%; total daytime charge at or below 30.69 US dollars: 3% to 6%	0.54	1.71		1997, 2003	agrees; AUC gain 0.005 [-0.0004, 0.012]	yes
C085	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by state: Washington minus state other than Washington; stronger in state other than Washington *shape: hinge at 223 minutes; 51% of patients same; fallback; risk saturates near 0%*	Washington: 11% to 14%; state other than Washington: 3% to 8%	-0.75	0.47		72, 3928	agrees; AUC gain 0.0004 [2e-05, 0.0008]	yes
C086	interaction: effect of voice-mail messages (0 messages to 32 messages) by total daytime charge: total daytime charge above 30.69 US dollars minus total daytime charge at or below 30.69 US dollars; stronger in total daytime charge at or below 30.69 US dollars *shape: steady; 100% of patients same; verified; risk saturates near 0%*	total daytime charge above 30.69 US dollars: 3% to 11%; total daytime charge at or below 30.69 US dollars: 3% to 11%	-0.53	0.59		1997, 2003	agrees; AUC gain 0.005 [-0.0004, 0.012]	yes
C087	interaction: effect of voice-mail messages (0 messages to 32 messages) by total daytime minutes: total daytime minutes above 180 minutes minus total daytime minutes at or below 180 minutes; stronger in total daytime minutes at or below 180 minutes *shape: steady; 100% of patients same; verified; risk saturates near 0%*	total daytime minutes above 180 minutes: 3% to 11%; total daytime minutes at or below 180 minutes: 3% to 11%	-0.53	0.59		1997, 2003	agrees; AUC gain 0.036 [0.015, 0.059]	yes
C088	threshold: total daytime minutes: fold at about 221 minutes (starts) *shape: hinge at 221 minutes; 48% of patients same; fallback; risk saturates near 0%; one point with C088 (median fold 221 minutes, typical patient 223 minutes, 10th-90th percentile 186 minutes to 221 minutes, wide)*	3% to 19% (typical patient with a fold)	1.98	7.25	1.48 to 2.79	3995, 1099	agrees; AUC gain 0.036 [0.015, 0.059]	yes
C089	threshold: voice-mail messages: fold at about 11 messages (stops) for customers in New Jersey *shape: hinge at 11 messages; 74% of patients same; fallback; one point with C089 (median fold 11 messages, typical patient 11 messages, 10th-90th percentile 4 messages to 27 messages, wide)*	24% to 13% (typical patient with a fold (customers in New Jersey))	-0.93	0.40	-1.97 to -0.27	36, 26	agrees; AUC gain 0.006 [0.003, 0.009]	yes
C090	threshold: total daytime charge: fold at about 39.34 US dollars (starts) for customers in North Carolina *shape: hinge at 39.34 US dollars; 88% of patients same; fallback; one point with C090 (median fold 39.34 US dollars, typical patient 39.00 US dollars, 10th-90th percentile 26.14 US dollars to 52.75 US dollars, wide)*	4% to 5% (typical patient with a fold (customers in North Carolina))	0.51	1.67	0.00 to 1.28	50, 17 (small n)	unclear; AUC gain 0.005 [-0.0004, 0.012]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C091	subgroup: who this part acts on: 1099 of 4000 patients	3% to 12% (typical patient this part acts on (risk without the part, then with it))	1.49	4.42	0.12 to 3.47	1099, 2901	agrees; AUC gain 0.080 [0.048, 0.112]	yes
C092	structure: its formula reproduces it on 4000 training patients (largest difference 4.44089e-16)		4e-16			4000	train only	yes
C113	boundary: boundary		0.00			4000	train only	no

T.7 State units

Annotation ann_85 (state_offsets). Reads: state.

Original hypothesis (unchanged). 1295 = relu(0.87 - 1.27 state:39) -> 7872 = relu(0.01 + 1.02*1295) (logit weights -1.01, -1.21); 9954 = relu(-0.38 + 1.01 state:15) (weight +0.04). Each depends only on one binary state indicator, so each is exactly an offset (least-squares R2 = 1.0). Hypothesis: the pair is a constant -2.33 logit for everyone except state:39 customers, for whom it is switched off (+2.36 logit), almost cancelling state:39's direct -2.61 (net -0.25, against an empirical log-odds offset of -0.93 for its 82 customers); 9954 adds +0.27 for state:15. Evidence: sensitivity 1.24/1.27 for state:39, 0.02 for state:15; linear replacement AUC 0.8908 (unchanged); mean ablation 0.8898.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 0.0).

$$
\begin{aligned}
y_0 &= \operatorname{relu}\big(0.868 - 1.27 x_{state:39_b}\big) \\
y_1 &= \operatorname{relu}\big(0.0138 + 1.02 \operatorname{relu}\big(0.868 \\
&\qquad - 1.27 x_{state:39_b}\big)\big) \\
y_2 &= \operatorname{relu}\big(-0.38 + 1.01 x_{state:15}\big)
\end{aligned}
$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C100	direction: state: state other than Indiana, Rhode Island to Rhode Island (through this part) *shape: categorical; highest to lowest: Rhode Island, Indiana, state other than Indiana, Rhode Island*	3% to 22%	2.36	10.6	2.36 to 2.36	3836, 82	unclear; AUC gain 0.005 [-0.002, 0.016]	yes
C101	direction: state: state other than Indiana, Rhode Island to Indiana (through this part) *shape: categorical; highest to lowest: Rhode Island, Indiana, state other than Indiana, Rhode Island*	3% to 3%	0.27	1.31	0.27 to 0.27	3836, 82	unclear; AUC gain 0.005 [-0.002, 0.016]	yes
C102	structure: its formula reproduces it on 4000 training patients (largest difference 0.0)		0.00			4000	train only	yes

T.8 Evening unit

Annotation ann_83 (evening_hinge). Reads: total evening minutes.

Original hypothesis (unchanged). 14484 = relu(-0.33 + 1.01 eveMinutes), 15524 = relu(0.71 + 1.02*14484) (always active); logit weight +0.20. Hypothesis: a small hinge in evening minutes, constant (+0.14 logit) below about 217 evening minutes and rising steadily above (+0.21 at 220-250, +0.36 at 250-300, +0.56 above 300); it adds to the direct evening-charge term (+0.35 per SD). Evidence: line plot correlation 0.79; mean ablation AUC 0.8907, linear replacement 0.8907 (R2 0.62): negligible for ranking.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 0.0).

$$y_0 = \mathrm{relu}\big(0.714 + 1.02\,\mathrm{relu}\big(-0.33 + 1.01 x_{totalEveMinutes}\big)\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C103	direction: total evening minutes: 138 minutes to 266 minutes (through this part) *shape: hinge at 217 minutes; 100% of patients same; verified; risk saturates near 0%; one point with C104 (median fold 217 minutes, typical patient 218 minutes, 10th-90th percentile 217 minutes to 217 minutes)*	3% to 3%	0.20	1.22	0.20 to 0.20	4000	unclear; AUC gain 0.003 [-8e-05, 0.005]	yes
C104	threshold: total evening minutes: fold at about 217 minutes (starts) *shape: hinge at 217 minutes; 100% of patients same; verified; risk saturates near 0%; one point with C104 (median fold 217 minutes, typical patient 218 minutes, 10th-90th percentile 217 minutes to 217 minutes)*	3% to 3% (typical patient with a fold)	0.28	1.32	0.28 to 0.28	4000, 1467	unclear; AUC gain 0.003 [-8e-05, 0.005]	yes
C105	structure: its formula reproduces it on 4000 training patients (largest difference 0.0)		0.00			4000	train only	yes

T.9 Small voice-mail unit

Annotation ann_84 (vmail_state18_unit). Reads: state, number of voice-mail messages.

Original hypothesis (unchanged). 6089 = relu(-0.67 + 1.14 state:18 + 0.83 vmailMessages), logit weight +0.005. On for customers with more than about 21 voicemail messages and for all 65 state:18 customers; logit contribution at most +0.05. Hypothesis: negligible. Evidence: mean ablation and linear replacement both leave train AUC at 0.8908.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 1.77636e-15).

$$y_0 = \mathrm{relu}\big(-0.668 + 1.14 x_{state:18_b} + 0.828 x_{numberVmailMessages_c}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C106	direction: state: state other than Louisiana to Louisiana (through this part) *shape: categorical; highest to lowest: Louisiana, state other than Louisiana*	3% to 3%	0.04	1.04	0.04 to 0.05	3935, 65	disagrees; AUC gain -6e-05 [-0.0002, 5e-05]	yes
C107	direction: voice-mail messages: 0 messages to 32 messages (through this part) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	3% to 3%	0.004	1.00	0.003 to 0.003	4000	unclear; AUC gain 2e-05 [-9e-05, 0.0001]	yes
C108	interaction: effect of voice-mail messages (0 messages to 32 messages) by state: Louisiana minus state other than Louisiana; stronger in Louisiana *shape: steady; 78% of patients same; fallback*	Louisiana: 5% to 19%; state other than Louisiana: 3% to 11%	0.14	1.15		65, 3935	unclear; AUC gain -6e-05 [-0.0002, 5e-05]	yes
C109	threshold: voice-mail messages: fold at about 22 messages (starts) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	3% to 3% (typical patient with a fold)	0.005	1.00	0.005 to 0.005	3935, 876	unclear; AUC gain 2e-05 [-9e-05, 0.0001]	no
C110	subgroup: who this part acts on: 941 of 4000 patients	2% to 2% (typical patient this part acts on (risk without the part, then with it))	0.006	1.01	0.0009 to 0.007	941, 3059	disagrees; AUC gain -5e-05 [-0.0002, 9e-05]	no
C111	null: this part barely changes predictions (switched off)		0.01			4000	train only	yes
C112	structure: its formula reproduces it on 4000 training patients (largest difference 1.77636e-15)		2e-15			4000	train only	yes

The network before any explanation

Figure 53 (next page, landscape) is the trained network as it is, with no parts, no splits and no inserted units: inputs at the bottom, hidden units by depth, the output at the top, every connection drawn, every input column its own node (one-hot levels included).

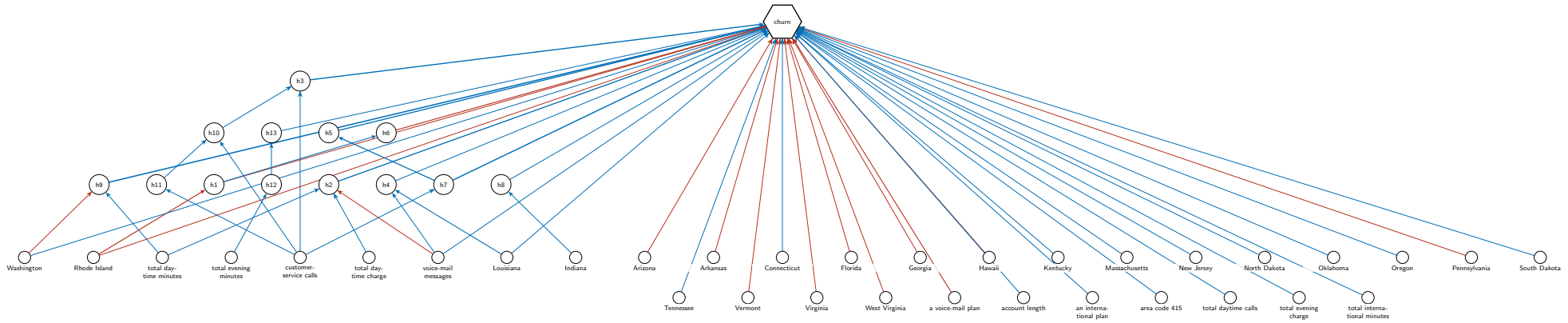


Figure 53: The network as trained, before any explanation, in full. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. Not used (no connection): Alaska, Alabama, Iowa, Idaho, Illinois, Kansas, Maryland, Maine, Michigan, Minnesota, Missouri, Mississippi, Montana, North Carolina, Nebraska, New Hampshire, New Mexico, Nevada, New York, Ohio, California, South Carolina, Texas, Utah, Wisconsin, Colorado, Wyoming, Washington DC, Delaware, area code 408, area code 510, total evening calls, total night-time minutes, total night-time calls, total night-time charge, total international calls, total international charge.